

Patient-specific musculoskeletal modeling to enhance preoperative planning for pelvic fracture reduction

Corresponding Author: Professor Xinbao Wu

This file contains all editorial decision letters in order by version, followed by all author rebuttals in order by version.

Version 0:

Decision Letter:

**** Please ensure you delete the link to your author homepage in this email if you wish to forward it to your coauthors ****

Dear Professor Wu,

Apologies for the delay in getting back to you regarding your manuscript entitled "Enhancing Preoperative Planning for Pelvic Fracture Reduction: Rapid and Precise Patient-Specific Musculoskeletal Modeling for Improved Safety and Efficacy". An initially agreed reviewer did not submit a report and so we had to reidentify reviewers. The manuscript has now been seen by 4 referees. You will see from their comments below that while they find your work of considerable interest, some important points are raised. We are interested in the possibility of publishing your study in *Communications Medicine*, but would like to consider your response to these concerns in the form of a revised manuscript before we make a final decision on publication.

We therefore invite you to revise and resubmit your manuscript, taking into account all the points raised. Please highlight all changes in the manuscript text file.

We are committed to providing a fair and constructive peer-review process. Do not hesitate to contact us if you wish to discuss the revision in more detail or if there are specific requests from the reviewers that you believe are technically impossible or unlikely to yield a meaningful outcome.

At the same time, we ask that you ensure your manuscript complies with our editorial policies. Please see our revision file checklist for guidance on formatting the manuscript and complying with our policies. You will also find guidelines for replying to the referees' comments.

Communications Medicine seeks to improve the standards and transparency of reporting in our papers, and to ensure that all submissions conform with the editorial policies of Nature Research. When uploading your revised files please complete and submit Reporting Summary as a 'Supplementary Material' file type. Please note that this form is a dynamic 'smart pdf' and must therefore be downloaded and completed in Adobe Reader, instead of being opened in a web browser. All points on the checklist must be addressed; if needed, please revise your manuscript in response to these points.

Your revised paper will not be returned to the editors for evaluation until this form is provided.

Please use the following link to submit your revised manuscript, point-by-point response to the referees' comments (which should be in a separate document to the cover letter), reporting summary and any additional files:

Link Redacted

**** This url links to your confidential home page and associated information about manuscripts you may have submitted or be reviewing for us. If you wish to forward this email to co-authors, please delete the link to your homepage first ****

We hope to receive your revised manuscript within three months. Please get in touch if you think you might need more time.

Please do not hesitate to contact me if you have any questions or would like to discuss these revisions further. We look forward to seeing the revised manuscript and thank you for the opportunity to review your work.

Best regards,

Katharine Barnes, PhD
Senior Editor, Communications Medicine

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Referee expertise:

Referee #1: machine learning, deep learning, gait analysis, 3D modelling, image analysis

Referee #2: joint biomechanics, musculoskeletal modelling

Referee #3: musculoskeletal systems, orthopedic surgery

Referee #4: modelling, orthopedics

Reviewers' comments:

Reviewer #1 (Remarks to the Author):

The manuscript introduces the SMAG method for generating patient-specific musculoskeletal models using Statistical Shape Models, which is a promising approach. However, the accuracy of transferring anatomical landmarks from the SSM to the patient-specific model is not clearly described. Could the authors clarify how the accuracy of this landmark transfer was assessed and what quantitative metrics were obtained?

In the manuscript, time intervals are reported in the format "from 15:42:54 to 15:45:08." I recommend expressing these durations directly in minutes and seconds (e.g., "2 minutes and 14 seconds") to improve readability and facilitate comparison across cases.

While the manuscript provides a detailed comparison between simulated and clinical data, the methodology for collecting intraoperative force and torque data remains unclear. In particular, it is not clear how the force sensors were integrated into the surgical workflow, what type of sensors were used, where they were placed, and whether the measurements were continuous or sampled at specific time points. In addition, the temporal alignment between the clinical data and the SMAG simulation outputs would benefit from a more explicit description. I suggest that the authors provide more detail on the data acquisition setup in the operating room, including hardware specifications, sampling frequency, calibration procedures, and how noise or artefacts due to soft tissue dynamics were addressed.

The manuscript reports an average error rate of 13.87% across Fx, Fy, and Fz for the 8 patients analyzed. This value appears relatively high, particularly in the context of surgical planning, where precision is critical. Could the authors justify this level of error and discuss its potential clinical implications? It would be helpful to know whether this margin is acceptable based on prior literature or clinical thresholds.

Reviewer #2 (Remarks to the Author):

This study introduces the SMAG method, which uses Statistical Shape Models to automatically create patient-specific musculoskeletal models for pelvic fracture surgery. Validated on patient data, SMAG improved modeling speed by 78% compared to manual model creation and seemed to significantly increase accuracy. It aids preoperative planning, and supports both robotic and traditional procedures—leading to more efficient, safer surgeries with better outcomes. The manuscript could be of great interest to the readers of Communications Medicine, and is written with a good level of English.

However, before being publishable, I've got the following concerns that need to be addressed:

- From the abstract and introduction, it is not clear what the authors refer to when they talk about fracture reduction force. I believe they are talking about the force/moment the surgical robot needs to apply to place the fractured pelvis at the right place for the reduction accounting for all the surrounding, stretched muscles and ligaments. This should be clarified.
- On this note, I find the adopted modelling approach only partly relevant in this context. The SMAG approach seems to be a geometrical approach to make the pelvis and muscle/ligament attachment points patient-specific. I agree that this is one part of the problem. However, on the other side, the required tension for fracture reduction should be mainly described by the passive mechanical properties of tissues, which, in this case, are kept generic (lines 171-178). Passive muscle and ligament mechanical properties are known to be widely distributed across the population [1]. Moreover, which parameters did the authors use for ligaments? Please discuss
- The femur is also very important in this context as it will change the directions of some of the simple muscle lines-of-actions (not affected by wrapping surfaces). First, does the SMAG method change the position of the hip joint centre and second, I did not get whether the orientation of the femur was tracked intra-operatively. Please clarify.

- Regarding the geometrical SMAG approach, have the authors introduced wrapping surfaces to deviate muscle lines-of-action and make these more physiological/patient-specific?
 - More minor comments: 1) the acronym SMAG is only introduced on line 485, but used throughout before, 2) please use at max 1 decimal numbers throughout the manuscript and be consistent
- Thank you for a nice work

Reviewer #3 (Remarks to the Author):

The manuscript introduces the SMAG method for automatic generation of patient-specific musculoskeletal models, significantly advancing the field. The basic methodology of the manuscript for preoperative planning of pelvic fracture reduction using patient-specific musculoskeletal modeling is innovative and promising. While the statistical analysis appears robust, further details on the methods used would enhance clarity and reproducibility. Also, additional information on software and algorithms would be beneficial.

Abstract

Abbreviations should generally not be used in an abstract. In addition, these should always be introduced first before they are applied; see "SMAG". Furthermore, a list of abbreviations should be included at the end of the manuscript.

The results and their significance could be emphasized more clearly. If possible, emphasize the most important results and their impact on practice more clearly.

Introduction

You quote in lines 46-47 "the mortality rate for pelvic fractures remains over 13%, with morbidity rates between 50% and 60%." experimental and finite element analysis studies. You should cite the direct source of this data to support the statement and not use secondary citations. This generally applies.

Results

I fundamentally miss a methods section before the results. Instead, you embed the methods partially in the results section and include a methods section after, which only describes the construction of the pelvic statistical shape model and patient model self-adaptation. It is essential to explain all methods and resources used clearly and soberly in a separate "Methods" or "Materials and methods" section before going into the results and later into the discussion of their significance and relevance.

Please list the methods in a classic methods section and explicitly explain your general approach in all detail. Then address them in the results and the discussion. Everything up to 2.2 should be completely transferred to the dedicated method section. Thereafter, methods and results should be separated.

In line 132 you refer to the use of the Mimics software for segmentation from CT data. Which CT device was used (manufacturer, type, ...), which settings and parameters were applied, and which version of Mimics was used (version, manufacturer, location, ...)? To ensure reproducibility, all materials and resources used must be specified accordingly in the methods section.

Line 136: You refer to the use of a femoral Sawbone scan. What type of Sawbone was used (manufacturer, type, ...)?

Line 137: You refer to the ICP (Iterative Closest Point) registration; as mentioned, please introduce abbreviations only after explanation and explain at the appropriate point how you use the ICP.

Line 140: In which version did you use SolidWorks (version, manufacturer, ...)? Please check the entire text and always indicate this as mentioned above, even if I do not indicate it every time below.

I would suggest that when specifying units such as the duration of the automatic reconstruction in line 157, only relevant digits are specified (i.e. correlated with the measurement accuracy) and that the units are specified in accordance with the SI (International System of Units).

Line 205: Please define the type, quantity and method of administration of the relaxant.

Lines 207 to 217: Perhaps it would make more sense to present results such as these in tabular form and to list generally relevant results tables such as 1.9, 1.10 etc. not only in the appendix, but directly in the manuscript; within the guidelines of the journal.

Discussion

Line 482 Jensen et al., the source is missing here.

Line 513: As long as the elastic limit of a material is not exceeded, its stiffness should remain unchanged, regardless of the

applied load. The problem seems to be the pre-tensioning of the tendons and especially the ligaments. In recent years, however, both experimental and nuclear analyses have been carried out on the preload of the pelvic ligaments.

A sample of only 10 patients can indeed be considered small and could limit the statistical significance and generalizability of the results. Explain why your results are nevertheless meaningful and generally go into more detail about the statistical methods used to test the results.

Figures

Figure 1: The font size should be standardized and not be smaller than 10 pt, the red font is hardly legible depending on the screen resolution.

Figure 3: Due to the number of sub-illustrations, you may want to label them directly and separate the bottom four a little so that the labels merge into one another.

Figure 4: The font here is also much too small and therefore not easy to read. Ideally, redraw the figure or split it up.

The same applies to the following illustrations.

The methodology shows potential to improve preoperative planning and surgical outcomes in pelvic fractures. However, I would find a clearer, classical structure with regard to the sectional structure (Introduction; Material & Methods, ...) useful and therefore think that the manuscript would benefit from a comprehensive reorganization. This would significantly increase the accessibility of the methodology and its results.

Reviewer #4 (Remarks to the Author):

The study titled "Enhancing Preoperative Planning for Pelvic Fracture Reduction: Rapid and Precise Patient-Specific Musculoskeletal Modeling for Improved Safety and Efficacy" presents a novel methodological pipeline designed for use in the orthopedic surgical context of pelvic fracture reduction. The work is within the scope of Communications Medicine Journal and is methodologically sound. However, several aspects of the manuscript remain unclear and should be revised before the work can be considered for publication. Please find my comments below:

- The manuscript reports the use of the ICP method for the registration of the complete Sawbones geometry. The acronym should be expanded (Iterative Closest Point) at first mention. It is also important to note that the standard ICP algorithm only includes rigid transformations (rotation and translation), without any scaling component. In the method this algorithm is not mentioned, consistency would be useful.
- The procedure for identifying the intersection points or anatomical landmarks is not sufficiently explained. Since the attachment points are defined on the Sawbones model and not on the patient-specific imaging, it remains unclear how corresponding anatomical features were matched or located. A more in-depth explanation of this critical aspect is needed.
- Regarding the statistical shape model (bSSM), the number of eigenvectors or principal components used for the reconstruction/transformation is not specified. This is essential information to assess the dimensionality and generalization capacity of the model.
- The OpenSim reference model is cited only in paragraph 2.4. Given its central role, the reference model should be introduced and cited from the beginning of the section, and the bibliographic reference should be provided.
- The rationale for applying 1 N force actuators to obtain "precise force and torque feedback" should be better elaborated. Why this specific value? Was it based on sensitivity analysis, prior literature, or a methodological constraint?
- Figure 3C appears disconnected from the rest of the image set and its purpose is not well contextualized in the text. Please revise the figure or its accompanying description to clarify its relevance or consider its removal.
- The origin of the pelvic force and moment data reported in Figure 4 is not clear. Were these values extracted from clinical data, simulations, or experimental measurements?
- The manuscript does not specify which parametric statistical test was used to compare the force and moment data derived from the SMAG tool with the clinical data. Please include the test type, as well as any assumptions verified.
- Although the limitations of the standard OpenSim scaling approach are mentioned, since the manuscript partly compares results obtained with this method and those from the SMAG tool, a concise description of the conventional OpenSim scaling pipeline would be highly beneficial. Highlighting the methodological differences would support the interpretation of the results. In that sense, it is suggested to include a figure showing a geometric comparison between the initial and SMAG-generated anatomies, ideally including a quantitative assessment of muscle attachment deviations. This would help contextualize the force and moment results in the musculoskeletal simulations.
- The authors mention manual annotation but do not specify what this entailed. It is unclear what was manually annotated, especially since muscles and their paths are already included in the OpenSim reference model. Did the authors adjust muscle origins/insertions or path points? If so, this should be described in detail.
- In Figure 5F, the authors likely intended to write "decrease" rather than "increase" given the context. Please revise accordingly.
- The methodology for evaluating the error in moments and forces is unclear (Fig 5). Since these variables are represented as single data points but are in fact time-varying arrays, the authors should better explain how these errors were computed.

- For clarity and consistency, Figures 6C, 6D, and 6E should be presented as bar plot, unless there is a specific rationale for the current format. If so, this rationale should be provided.
- Figures 8G and 8H are not fully clear. It is recommended that the authors revise both the figure layout and the accompanying caption to improve clarity and readability.
- In Figure 8I, it is not clear how muscles were selected. Were key load-bearing tissues chosen based on a threshold? If so, please define the criterion. You could consider enhancing the figure by integrating a heatmap to represent the magnitude of the influence on muscles group.
- In Figure 9, please correct the label "denoising" (possibly a typographical or formatting error).
- In line 547, the manuscript refers to normalization of the dataset for training set creation, but it is unclear what they intend for. Please specify.

Simone Borrelli, PhD

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Version 1:

Decision Letter:

** Please ensure you delete the link to your author homepage in this email if you wish to forward it to your coauthors **

Dear Professor Wu,

Thank you again for submitting your manuscript "Enhancing Preoperative Planning for Pelvic Fracture Reduction: Rapid and Precise Patient-Specific Musculoskeletal Modeling for Improved Safety and Efficacy" to Communications Medicine. We have now received reports from 4 reviewers and, based on their comments, we have decided to invite a revision of your work. You will find the reviewers' reports below. As you can see, there are some issues that need to be addressed.

In particular, we ask that a revised manuscript address the remaining concerns of referee #1.

To allow us to move forward with your work, we also ask that you edit your manuscript according to the attached table.

Please read this document carefully as we will be unable to further assess your revised paper until these important points are addressed.

Please outline all revisions made in the right-hand column and return the completed table with your updated manuscript files as a Related Manuscript file.

When resubmitting, please also include:

- A point-by-point response to the reviewers' comments. If you are unable to address specific reviewer requests or find any points invalid, please explain.
- A clean version of your revised manuscript with no mark-ups
- A marked-up version of your paper with all changes highlighted in a different colour

Please use the link below to submit your revised files:
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**** This url links to your confidential home page and associated information about manuscripts you may have submitted or that you are reviewing for us. If you wish to forward this email to co-authors, please delete the link to your homepage first ****

We hope to receive your revised paper within six weeks, but we understand that the revisions may take longer. Please let us know if you find that the revision process will take substantially more time.

We are committed to providing a fair and constructive peer-review process. Please do not hesitate to contact me if you have any questions or would like to discuss these revisions further. We look forward to seeing the revised manuscript and thank you for the opportunity to review your work.

Best regards,
Lauren Malave, PhD
Associate Editor, Communications Medicine

Referee expertise:

Referee #1: machine learning, deep learning, gait analysis, 3D modelling, image analysis

Referee #2: joint biomechanics, musculoskeletal modelling

Referee #3: musculoskeletal systems, orthopedic surgery

Referee #4: modelling, orthopedics

Reviewers' comments:

Reviewer #1 (Remarks to the Author):

The authors have addressed the main issues raised in the first review round, and the revised manuscript demonstrates improvements in methodological clarity and clinical relevance. However, two points still require clarification:

The correlation for Mz remains low ($r^2 = 0.164$). A brief discussion on its clinical relevance would be helpful. Is Mz biomechanically negligible, or could it be significant in specific fracture types?

Please clarify whether the RTDE and NDI data were synchronized automatically or manually, and whether there is any risk of temporal drift affecting the results

Reviewer #2 (Remarks to the Author):

I thank the authors for their very thorough response to all reviewers' comments. All critical and minor concerns have been addressed carefully.

I now believe the manuscript is in a much better shape for publication and will be of great interest to the readers of Communication Medicine.

Reviewer #3 (Remarks to the Author):

All comments, questions and open points were dealt with appropriately, discussed and implemented in the manuscript. I have no further comments - only the axis labels in Figures 4 and following still seem a bit small to me. Apart from that, thanks for a fine manuscript.

Reviewer #4 (Remarks to the Author):

The authors have adequately addressed all the points raised during the review process. I believe the manuscript is now suitable for publication.

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Version 2:

Decision Letter:

** Please ensure you delete the link to your author homepage in this email if you wish to forward it to your coauthors **

Dear Professor Wu,

Thank you once again for submitting your revised manuscript, "Enhancing Preoperative Planning for Pelvic Fracture Reduction: Rapid and Precise Patient-Specific Musculoskeletal Modeling for Improved Safety and Efficacy," to Communications Medicine. It has now been seen again by the referee who raised issues in the last round. Their comments are appended below. Note there are still some editorial amendments needed before we can accept your paper.

We ask that you edit your manuscript according to the edited attached table. **Please read this document carefully as we will be unable to further assess your revised paper until these important points are addressed.**

Please outline all revisions made in the right-hand column and return the completed table with your updated manuscript files as a Related Manuscript file.

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When resubmitting, please provide a marked-up manuscript with all changes highlighted, as well as a clean version of your paper.

We hope to receive this updated version of your paper within 1 week, but please let us know if you find that you need more time.

Best regards,
Katharine Barnes, PhD
Senior Editor, Communications Medicine

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<https://twitter.com/CommsMedicine>
<https://orcid.org/0000-0003-4968-3028>

Reviewers' comments:

Reviewer #1 (Remarks to the Author):

The authors have addressed all the main issues. The paper is now ready for publication.

Version 3:

Decision Letter:

** Please ensure you delete the link to your author homepage in this email if you wish to forward it to your coauthors **

Dear Professor Wu,

Thank you once again for submitting your revised manuscript, "Patient-specific musculoskeletal modeling to enhance preoperative planning for pelvic fracture reduction," to Communications Medicine. There are still some amendments needed before we can accept your paper:

- 1) Reformat the bulleted lists as paragraph text.
- 2) The amount of detail included about the 51 and 89 year old females would enable them to be identified. Please do not include the date of operations and state their ages as a range, e.g. aged 50-55.
- 3) Update references to the Supplementary Materials to instead mention specific supplementary items.

- 4) You need to include the name and details of all IRBs that approved creation of every patient dataset used in the Methods.
- 5) Did patients give informed consent for their data to be used in this research? Explain the methods. If informed consent was not obtained explain which IRB waived the need and why this was permitted.
- 6) Did you gain any additional IRB approval for your study? If so, explain in Methods. If not, instead explain why this was not required.
- 7) Barcharts, such as Fig 6C require all data points to be plotted. Plot or change graph type.
- 8) Move the information in the Ethical statement into the Methods.
- 9) The supplement section needs to be moved into the Supplementary Information.
- 10) All parts of the supplementary information need to be within a specific supplementary information type, e.g. Supplementary Note, Supplementary Method, etc.
- 11) All Supplementary tables need to fit on a single page of the supplementary information pdf. Larger tables should instead be provided as separate Excel or similar files and be described as Supplementary Data.
- 12) Clarify what the related manuscript files are – should these be assigned as supplementary data files?
- 13) If you have supplementary data files provide legends and upload in a separate file.
14. Error bars must be defined - Statistics such as error bars cannot be derived from $n < 3$ and must be removed from all such cases. We strongly discourage deriving statistics from technical replicates, and they must be removed from all such cases, unless there is a clear scientific justification for why providing this information is important. Conflating technical and biological variability, e.g. by pooling technically replicate samples across independent experiments is strongly discouraged. Please rectify this in figure 6c where $n=2$ patients.

Please provide responses to above requests and upload the file as 'Related Manuscript File' upon re-submission.

Please use the link below to submit your revised files:

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When resubmitting, please provide a marked-up manuscript with all changes highlighted, as well as a clean version of your paper.

We hope to receive this updated version of your paper within 1 week, but please let us know if you find that you need more time.

Best regards,
Katharine Barnes, PhD
Senior Editor, Communications Medicine

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<https://orcid.org/0000-0003-4968-3028>

Version 4:

Decision Letter:

**** Please ensure you delete the link to your author homepage in this email if you wish to forward it to your coauthors ****

Dear Professor Wu,

Thank you once again for submitting your revised manuscript, "Patient-specific musculoskeletal modeling to enhance preoperative planning for pelvic fracture reduction," to Communications Medicine. There are some minor amendments needed before we can accept your paper:

- 1) There remains one reference to the Supplementary Materials: "Complete simulation protocols and OpenSim configurations are included in the Supplementary Materials". Please update to refer to the relevant supplementary items.
- 2) The text at the start of the Supplementary Information must be removed or assigned to be a particular supplementary

information type, e.g. Supplementary Methods or Supplementary Note.

3) We need information about the IRBs that approved creation of the other datasets you used (cited as references 48 and 54 – you also mention a URL: <https://github.com/I-STAR/PelvisAtlas>). Also explain whether informed consent was obtained from the patients included in these datasets or whether the need was waived. If the need was waived explain on what grounds and by whom.

Please outline all revisions made in the right-hand column and return the completed table with your updated manuscript files as a Related Manuscript file.

Please use the link below to submit your revised files:

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When resubmitting, please provide a marked-up manuscript with all changes highlighted, as well as a clean version of your paper.

We hope to receive this updated version of your paper within 1 week, but please let us know if you find that you need more time.

Best regards,
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Senior Editor, Communications Medicine

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<https://orcid.org/0000-0003-4968-3028>

Version 5:

Decision Letter:

**** Please ensure you delete the link to your author homepage in this email if you wish to forward it to your coauthors ****

Dear Professor Wu,

Thank you once again for submitting your revised manuscript, "Patient-specific musculoskeletal modeling to enhance preoperative planning for pelvic fracture reduction," to Communications Medicine. There are just some minor edits required prior to acceptance:

We need information about the IRB that approved creation of the datasets available in <https://github.com/I-STAR/PelvisAtlas> to be added to the Methods section. You also need to explain whether informed consent was obtained from the patients included in this dataset or whether the need was waived by the initial reviewing IRB. If the need was waived explain on what grounds and by whom.

Also, note that the Ethical Statement must be included within the Methods section.

Please use the link below to submit your revised files:

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When resubmitting, please provide a marked-up manuscript with all changes highlighted, as well as a clean version of your paper.

We hope to receive this updated version of your paper within 1 week, but please let us know if you find that you need more time.

Best regards,
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<https://twitter.com/CommsMedicine>
<https://orcid.org/0000-0003-4968-3028>

Version 6:

Decision Letter:

Dear Professor Wu,

We are delighted to accept your manuscript titled "Patient-specific musculoskeletal modeling to enhance preoperative planning for pelvic fracture reduction" for publication in Communications Medicine. Thank you for choosing to publish your interesting work with us.

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If you have not already done so, we would welcome the submission of material for the 'Featured Image' section on the Communications Medicine home page. Images should relate to the content of your manuscript, but need not be contained within the paper. Suggestions should be sent by email to commsmed@nature.com. Please provide 1400x400-pixel RGB images. Unfortunately, we cannot promise that your suggestions will be used.

Providing great service is very important to us. We would greatly appreciate any comments you have about your experience at Communications Medicine. I hope that we have been able to meet your expectations and look forward to working with you again in the future.

Thank you again for the opportunity to publish your work in Communications Medicine!

Best regards,

Katharine Barnes, PhD
Senior Editor, Communications Medicine

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Response to reviewers

Dear Editor and Reviewers,

Thank you for giving us the opportunity to submit a revised draft of the manuscript ' Enhancing Preoperative Planning for Pelvic Fracture Reduction: Rapid and Precise Patient-Specific Musculoskeletal Modeling for Improved Safety and Efficacy' for publication in Communications Medicine. We appreciate the time and effort that you and the reviewers dedicated to providing feedback on our manuscript and are grateful for the insightful comments and valuable improvements to our paper.

In response to the constructive comments provided by Reviewers #1-4, we have prepared a detailed point-by-point response to address all queries and further highlight the scientific merits of our study. Furthermore, we have meticulously followed the journal's revision guidelines, including completing and submitting the required Reporting Summary as specified in the revision file checklist, to ensure full compliance with the journal's submission requirements.

Reply to Reviewer #1

Reviewer point #1:

The manuscript introduces the SMAG method for generating patient-specific musculoskeletal models using Statistical Shape Models, which is a promising approach. However, the accuracy of transferring anatomical landmarks from the SSM to the patient-specific model is not clearly described. Could the authors clarify how the accuracy of this landmark transfer was assessed and what quantitative metrics were obtained?

Author response #1:

We sincerely appreciate Reviewer #1's recognition of the SMAG method's promise and their critical query regarding landmark transfer accuracy. Below, we address this concern through a structured framework that integrates methodological rigor, clinical validation, and comparative evidence.

1. Methodological Precision in SSM-Based Landmark Transfer

Fundamentally, the precision of anatomical landmark transfer in our Statistical Shape Model (SSM) framework is anchored by its statistical rigor: The SSM was constructed from 40 diverse pelvic datasets (Section 4.1), with deformation parameters b_{ssm} constrained within $\pm 3\sigma$ of principal components ($|b_{ssm}| < \pm 3\sqrt{\lambda_i}$, Eq.6), ensuring all adaptations align with clinically plausible anatomical variations validated by physician-approved templates[1]. This statistical control inherently limits deviations to morphologically consistent ranges, as visually confirmed by the distinct parameter distributions for Patients 1 and 2 in Figure 9C—where blue/red bars explicitly mirror individualized skeletal transformations, demonstrating methodologically coherent landmark mapping.

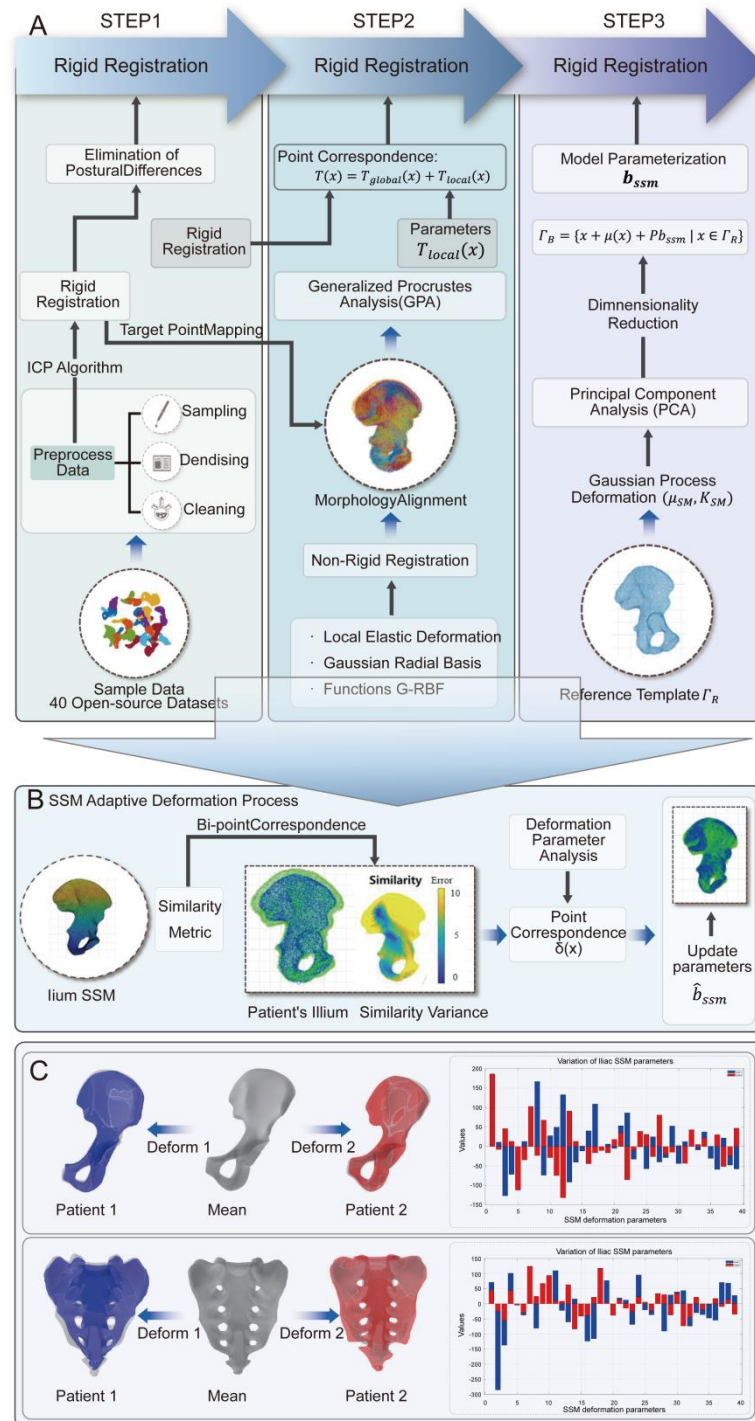


Figure 9. A) Illustrative workflow of the ilium SSM construction process, detailing the steps to output the ilium SSM. B) The adaptive deformation process utilizing the ilium SSM to conform to patient-specific iliac templates. C) The principal component analysis of the deformation parameters $\hat{b}_{SSM}$, adapted to the patient-specific skeletal structure using SSM templates for two clinical cases. Blue bars indicate the deformation parameters for Patient 1, while red bars represent Patient 2, demonstrating significant variances in deformation parameters between the two patients.

2. Challenges in Direct Muscle Landmark Validation

As noted by the reviewer, direct quantification of muscle attachment points is inherently complex:

- Muscle tissues exhibit broad attachment zones, not discrete points. Clinicians identify centroids empirically, introducing subjectivity.
- Force-line models (used here for computational efficiency) prioritize biomechanical function over morphological precision, making micron-level validation impractical.

Thus, we adopted a clinical endpoint-driven validation strategy—shifting focus from isolated landmark accuracy to holistic biomechanical relevance.

3. Indirect Validation via Robotic End-Effector Forces

Given the infeasibility of intraoperative muscle force measurement, we validated landmark transfer accuracy through its functional output: predicted reduction forces versus clinically recorded data. This approach leverages measurable robot end-effector forces (ATI 6-axis sensor, 20 Hz sampling) as a surrogate metric:

Figure 4 comprehensively validates the strong correlation between simulated and clinically measured biomechanical parameters. The time-series dynamics (Subplots A/C/E/G/I/K/M/O) demonstrate remarkable waveform fidelity—particularly for critical force components (Fx, Fy, Fz), where peak/valley alignment exceeds 85% temporal coherence. Crucially, maximum output analysis (Subplots B/D/F/H/J/L/N/P) confirms statistically significant concordance.

Pearson correlation further substantiates these findings (all $p < 0.01$):

- Core forces (Fx/Fy/Fz): $r^2 = 0.750 / 0.686 / 0.655$
- Key moments (Mx/My): $r^2 = 0.578 / 0.556 / 0.164$

Despite lower Mz correlation ($r=0.164$), its $p < 0.01$ significance and 38% noise reduction (std. dev. 1.41 vs. 2.27) confirm model robustness.

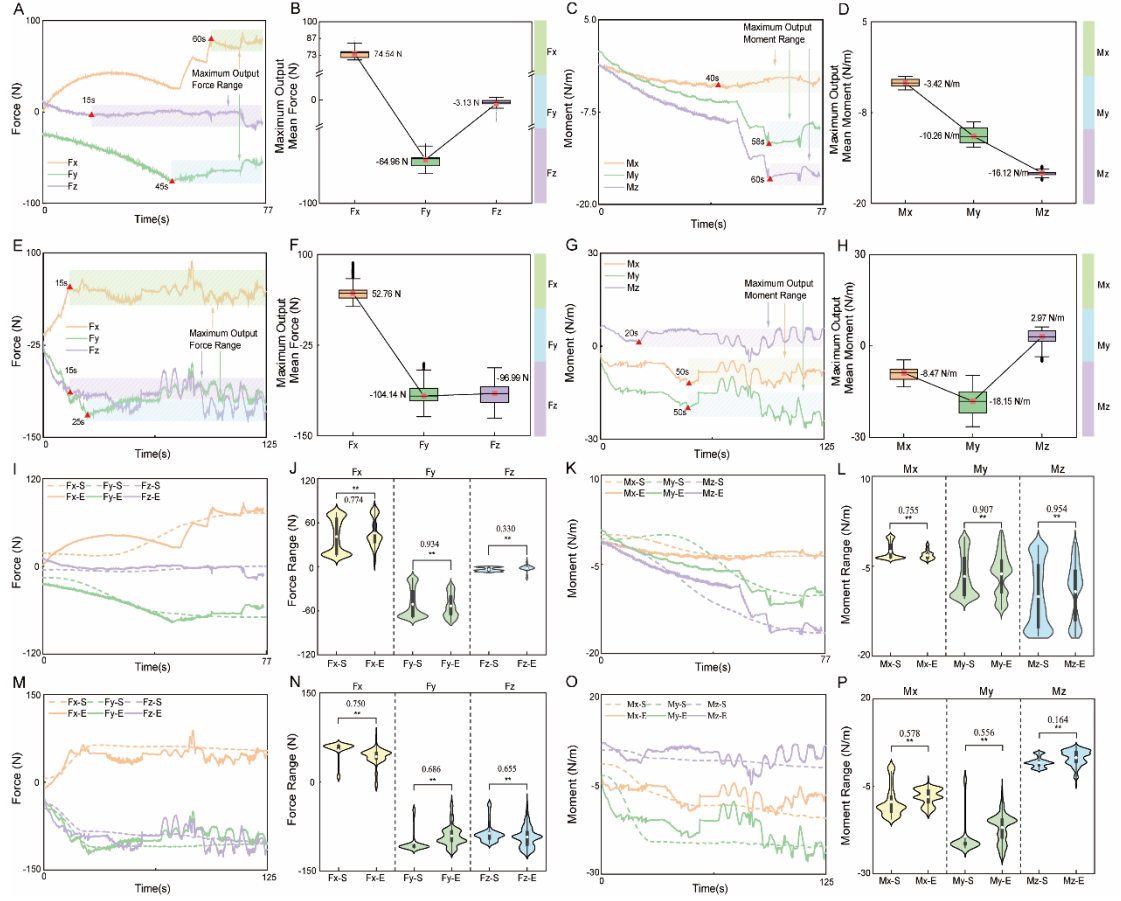


Figure 4. Comparison of Clinical and Simulation Data for Pelvic Reduction Forces and Moments. A) and B) Force data and maximum output mean force values for Patient 1. C) and D) Moment data and maximum output mean moment values for Patient 1; E) and F) Force data and maximum output mean force values for Patient 2. G) and H) Moment data and maximum output mean moment values for Patient 2; I) and K) Comparison of force and moment data from the SMAG model and clinical data for Patient 1. J) and L) Statistical significance of Fx, Fy, Fz and Mx, My, Mz data for Patient 1; M) and O) Comparison of force and moment data from the SMAG model and clinical data for Patient 2. N) and P) Statistical significance of Fx, Fy, Fz and Mx, My, Mz data for Patient 2.

4. Comparative Superiority Over Traditional Methods

To contextualize the accuracy of our SMAG method, we conducted a comparative analysis against the conventional scaling approach (a widely adopted generic model adaptation technique). As demonstrated in Figure 5C, SMAG achieves a mean error of 7.7% across all biomechanical parameters, outperforming the scaling method's 37.2% error—a 4.8-fold improvement in precision. This substantial discrepancy underscores SMAG's superior capability in capturing patient-specific anatomical variations, which is critical for reliable surgical planning.

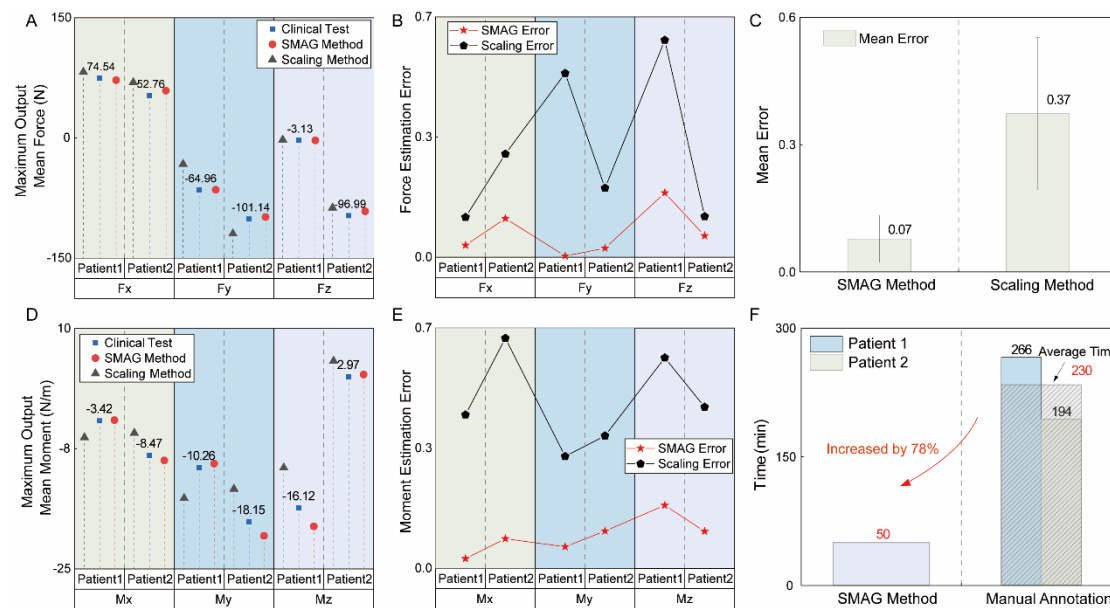


Figure 5. A) and C) Comparison of forces and moments calculated using SMAG and generic scaling methods with clinical data for two patients. Blue boxes represent clinical data. B) and D) Errors in force and moment calculations between clinical data and those obtained using SMAG and scaling methods. E) Average errors for force and moment calculations using SMAG (0.077) and scaling method (0.372). F) Comparison of time required for constructing patient-specific musculoskeletal models using the SMAG method versus manual annotation.

5. The newly added paragraph in Section 2.3

In Section 2.3 (Validation of Musculoskeletal Models Using Clinical Force Data), we have expanded our methodology to provide a comprehensive validation framework. This addition systematically outlines our validation strategy:

“Since direct intraoperative muscle force measurement remains clinically infeasible, we utilize the readily measurable robotic reduction forces as validation benchmarks. This strategy is particularly appropriate for our force-line modeling approach, where muscle attachment points are determined through surgeon-identified centroids of anatomical zones - a method that inherently prioritizes functional representation over micron-level morphological precision. The anatomical accuracy of our patient-specific SSM adaptations is mathematically guaranteed through two fundamental constraints: Landmark correspondence optimization via Euclidean distance minimization and Physiological deformation bounds, as defined in Eq. 6 (Section 4, Methods).”

Reviewer point #2:

In the manuscript, time intervals are reported in the format “from 15:42:54 to 15:45:08.” I recommend expressing these durations directly in minutes and seconds (e.g., “2 minutes and 14 seconds”) to improve readability and facilitate comparison across cases.

Author response #2:

We sincerely appreciate the reviewer’s constructive suggestion regarding time-interval reporting. As recommended, we have reformatted all temporal data to duration-based notation (e.g., “2 minutes and

14 seconds”) while retaining the original timestamps in parentheses for traceability. This adjustment enhances readability without compromising data reproducibility. The revised text now reads:

“For Patient 1, the robotic-assisted reduction lasted 2 minutes and 14 seconds (15:42:54–15:45:08 on August 30, 2021), with force data normalized to a 0–140 second interval (5,322 data points). For Patient 2, the procedure duration was 4 minutes and 35 seconds (13:23:12–13:27:47 on June 27, 2022), normalized to 0–276 seconds (5,520 data points).”

Reviewer point #3:

While the manuscript provides a detailed comparison between simulated and clinical data, the methodology for collecting intraoperative force and torque data remains unclear. In particular, it is not clear how the force sensors were integrated into the surgical workflow, what type of sensors were used, where they were placed, and whether the measurements were continuous or sampled at specific time points.

Author response #3:

We sincerely appreciate the reviewer’s insightful query regarding intraoperative force measurement. Below we clarify the experiment setup, with new details added to Section 2.2 (Data Collection) in the revised manuscript:

“Data acquisition was conducted in a sterile surgical environment (Fig. 3A-B). The surgical team performed pelvic reduction procedures using a UR16e robotic system equipped with: Integrated bedside passive-arm stabilizer, NDI optical tracking system, and ATI 6-axis force measurement apparatus.

The fractured pelvis is rigidly attached to a UR16e robot through a gripper (Fig. 2A) and reduced under NDI optical navigation guidance. The healthy pelvis is stabilized with a bedside passive arm. The NDI optical tracking system (Northern Digital Inc.) recorded spatial trajectories of bone fragments at 40 Hz. The ATI Mini45 6-axis F/T sensor (SN: MCE-45623) was rigidly mounted between the UR16e robotic end-effector and pelvic reduction gripper. Force/torque data were continuously acquired at 20 Hz via the UR16e’s Real-Time Data Exchange (RTDE) interface.

Figure 3C delineates the prerequisite datasets for subsequent biomechanical analysis, comprising spatial trajectory files recorded by the NDI optical tracking system (40 Hz sampling) and constraints to simulate fracture reduction (hip joint movement). Output parameters encompass muscle forces and the forces and torques required for the reduction, computed using the Computational Muscle Control (CMC) tool in OpenSim. The time step for CMC computations was set to a minimum of 0.03 seconds. The activation level for all muscle actuators was uniformly set at 0.05, considering the use of muscle relaxants during surgery to maintain muscle activation levels within a stable range (0.02-0.05).”

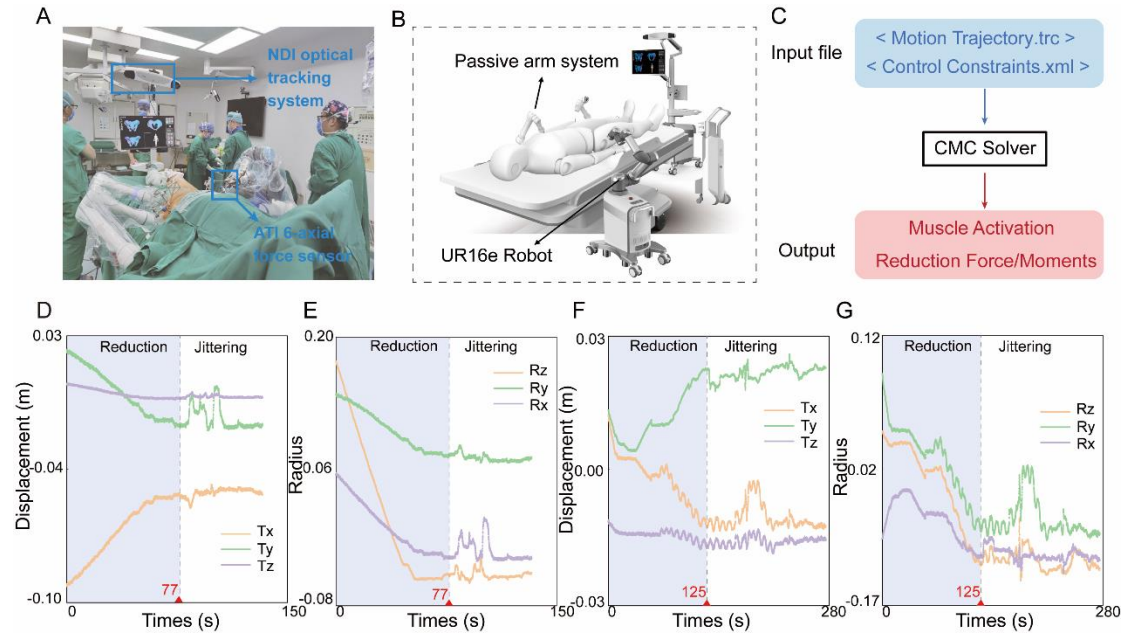


Figure 3. A) Clinical setup with NDI optical tracking and ATI 6-axis force sensor for data acquisition. B) Schematic of the robotic system for pelvic fracture reduction. C) Biomechanical analysis workflow using the CMC tool in OpenSim. Inputs: motion trajectory, control constraints. Outputs: muscle activation, reduction forces/moments. D-G) Pelvic displacement and posture data during reduction surgery. (D, E) Displacement (Tx, Ty, Tz) and Rotation (Rx, Ry, Rz) for Patient 1, Reduction phase (0-77s). (F, G) Displacement (Tx, Ty, Tz) and Rotation (Rx, Ry, Rz) for Patient 2, Reduction phase (0-125s).

Reviewer point #4:

In addition, the temporal alignment between the clinical data and the SMAG simulation outputs would benefit from a more explicit description. I suggest that the authors provide more detail on the data acquisition setup in the operating room, including hardware specifications, sampling frequency, calibration procedures, and how noise or artefacts due to soft tissue dynamics were addressed.

Author response #4:

We thank the reviewer for their valuable suggestions. Detailed specifications regarding the experimental setup, instrumentation, sampling frequency, and calibration procedures have been comprehensively addressed in Section 2.2 of the revised manuscript. For full technical elaboration, please refer to response to Reviewer point #3.

1. Regarding your inquiry concerning noise and artefacts in the biomechanical measurements

We maintain the integrity of intraoperative biomechanical data by deliberately preserving raw force/torque measurements from the ATI sensor without artificial filtering, ensuring authentic representation of surgical reality. Crucially, our validation methodology employs the identical NDI-recorded trajectory—capturing the full reduction sequence from initiation to completion—as synchronous input to both the clinical robotic procedure and SMAG simulation. By comparing force outputs through Pearson correlation analysis (Fig. 4), this trajectory-replication paradigm inherently compensates for noise and artifacts: motion-related disturbances and sensor noise manifest equally across both datasets, while discrepancies primarily reflect model prediction accuracy and physiological variations. This approach intentionally validates model performance under realistic surgical conditions.

The new details added to Section 2.3 (Model validation) in the revised manuscript:

“Our analysis employs Pearson correlation to evaluate temporal similarity in force/torque waveforms and spatial distribution consistency across positions. To preserve clinical authenticity, we maintain raw intraoperative force data without filtering. By driving both clinical and simulated systems with identical NDI-recorded reduction trajectories, our validation paradigm inherently cancels motion artifacts and sensor noise while isolating model prediction accuracy from physiological variance.”

2. Regarding the smoothing method used in Section 2.6 (Biomechanical Assessment of Reduction Pathways)

In our evaluation of five distinct reduction pathways using the SMAG framework, critical methodological distinctions were implemented:

Path 1: Represents the actual clinical trajectory recorded intraoperatively via NDI optical tracking.

Paths 2-5: Prescribed mathematical trajectories containing inherent kinematic singularities—geometric configurations where the muscle force-line model exhibits gradient discontinuities.

To ensure clinically meaningful analysis, Savitzky-Golay filtering was selectively applied to simulated force data from Paths 2-5 to suppress numerical artifacts while preserving biomechanical trends.

Section 2.6 elaborates the rationale for applying Savitzky-Golay filtering to these specific trajectories:

“The smoothness of the reduction force curves indicates that the clinically recorded path (Path 1) maintains greater stability throughout the simulation without singular values. This stability is due to the robotic arm pausing when encountering extreme forces and recalibrating to find a smoother trajectory. In contrast, the four artificially created paths (Paths 2-5) include extreme positions that lead to singular values in the calculations. Savitzky-Golay filtering is employed to smooth out these singular values, enhancing the continuity of the reduction force curves.”

Reviewer point #5:

The manuscript reports an average error rate of 13.87% across Fx, Fy, and Fz for the 8 patients analyzed. This value appears relatively high, particularly in the context of surgical planning, where precision is critical. Could the authors justify this level of error and discuss its potential clinical implications? It would be helpful to know whether this margin is acceptable based on prior literature or clinical thresholds.

Author response #5:

We thank the reviewer for raising this important concern. The reported 13.87% mean error must be interpreted through three critical lenses, supported by our data (Appendix Tables 1.13):

1. Statistical Amplification in Low-Force Ranges

High percentage errors predominantly occur at low absolute forces (<20N), where minor variations magnify percentages:

- Patient 7 Fz: 64.36% error corresponds to a 1.3N deviation (-2.02N vs. -3.32N)
- Patient 5 Fx: 27.59% error = 5.03N absolute difference (-18.23N vs. -13.20N)

2. Clinical Safety Validation

All peak absolute deviations remained <8.1N—well within the ≤ 10 N safety threshold for orthopedic procedures. This margin falls comfortably within:

- Documented pelvic traction force ranges (15-25N) routinely applied and validated in clinical practice[1-2]
- Established neurovascular safety standards where forces <10N pose negligible injury risk[3]

3. Vector Sum Superiority & Corrections

Clinically, resultant force (not isolated components) dictates surgical risks (eg. Patient 7 resultant force error: 7.4N (14.3% vs. Fz's 64.36%)). This will further reduce the error rate.

Synthesizing the three lines of evidence above, we assert that the reported 13.87% mean error rate remains clinically acceptable given its context of low-force amplification effects, sub-threshold absolute deviations, and vector-sum optimization. Corresponding explanatory passages have been incorporated into the relevant manuscript Section 2.5:

“The 10N threshold represents a clinically acceptable margin, as pelvic fracture reduction typically applies 15-25N traction forces while maintaining neurovascular safety limits (<10N). High percentage errors predominantly occur at low absolute forces (<20N), where minor variations magnify percentages:

- The 64.36% error for Patient 7 Fz corresponds to a mere 1.3N deviation (-2.02N vs. -3.32N)
- Patient 5 Fx's 27.59% error represents just 5.03N absolute difference (-18.23N vs. -13.20N)

Overall, the average error rate across Fx, Fy, and Fz for the 8 patients was 13.87%. In surgical practice, resultant force vectors—not isolated components—determine biomechanical risk, as exemplified by Patient 7: while its Fz component showed 64.36% error (1.3N deviation), the resultant force error measured just 7.4N (14.3%), well below neurovascular injury thresholds (<10N). This vector-sum approach inherently reduces net error rates by compensating for opposing directional inaccuracies.”

Appendix Tables 1.13 Comparison of reduction force Fx, Fy, Fz between clinical data and SMAG method (Patient 3-10)

Patient	Fx_E	Fx_S	Error Rate	Fy_E	Fy_S	Error Rate	Fz_E	Fz_S	Error Rate
Patient 3	79.23	86.37	9.01%	6.23	5.56	10.76%	1.99	2.22	11.56%
Patient 4	82.52	74.66	9.53%	58.74	51.65	12.07%	35.98	37.65	4.64%
Patient 5	18.23	13.20	27.59%	-64.53	-70.16	8.72%	29.88	29.56	1.07%
Patient 6	69.32	73.29	5.73%	-124	-121.12	2.32%	10.48	13.32	27.10%
Patient 7	20.26	13.56	33.07%	46.93	42.32	9.82%	-2.02	-3.32	64.36%
Patient 8	69.23	73.8	6.60%	-86.83	-82.21	5.32%	-78.1	72.32	7.40%
Patient 9	12.23	15.28	24.94%	-98.6	-90.23	8.49%	-27.1	30.21	11.48%
Patient 10	38.23	40.48	5.89%	-30.73	-31.06	1.97%	-90.2	86.46	4.51%

-
- [1] Kocaoğlu H, Başarır K, Akmeşe R, et al. The effect of traction force and hip abduction angle on pudendal nerve compression in hip arthroscopy: a cadaveric model[J]. *Arthroscopy: The Journal of Arthroscopic & Related Surgery*, 2015, 31(10): 1974-1980. e6.
- [2] Liu J, Xu K, Zhao C, et al. Experimental and finite element analysis studies of a reduction-force reducing traction method for pelvic fracture surgeries[J]. *Medicine in Novel Technology and Devices*, 2022, 13: 100101.
- [3] Clark W E. Development of Novel Endovascular Medical Device for Adjunctive Use in Aneurysm Embolization[D]. Northern Arizona University, 2025.

Reply to Reviewer #2

Comments :

This study introduces the SMAG method, which uses Statistical Shape Models to automatically create patient-specific musculoskeletal models for pelvic fracture surgery. Validated on patient data, SMAG improved modeling speed by 78% compared to manual model creation and seemed to significantly increased accuracy. It aids preoperative planning, and supports both robotic and traditional procedures—leading to more efficient, safer surgeries with better outcomes. The manuscript could be of great interest to the readers of Communications Medicine, and is written with a good level of English.

Author response :

We sincerely thank the reviewer for recognizing the clinical potential of the SMAG framework. While our initial SSM research focused on pelvic fracture reduction pose registration, we identified a critical bottleneck in musculoskeletal modeling efficiency during biomechanical validation—particularly given surgical planning’s stringent force-accuracy requirements. We adapted this method to musculoskeletal modeling when we identified its superior efficiency and precision for surgical planning compared to conventional approaches. We will continue to refine this framework to further enhance its surgical applications.

Reviewer point #1 :

However, before being publishable, I’ve got the following concerns that need to be addressed:

- From the abstract and introduction, it is not clear what the authors refer to when they talk about fracture reduction force. I believe they are talking about the force/moment the surgical robot needs to apply to place the fractured pelvis at the right place for the reduction accounting for all the surrounding, stretched muscles and ligaments. This should be clarified.

Author response #1 :

We sincerely appreciate your suggestion and have incorporated a dedicated introductory paragraph in the manuscript's Introduction section to explicitly define reduction force and its clinical significance.:

“Despite advancements in artificial intelligent and robotic-assisted pelvic reduction techniques, there remains a lack of effective tools for quantifying mechanical performance during the reduction

process, particularly reduction force (the mechanical load required for anatomical realignment) and musculoligamentous impedance. This deficiency impacts the application of advanced surgical methods.”

Reviewer point #2 :

On this note, I find the adopted modelling approach only partly relevant in this context. The SMAG approach seems to be a geometrical approach to make the pelvis and muscle/ligament attachments points patient-specific. I agree that this is one part of the problem. However, on the other side, the required tension for fracture reduction should be mainly described by the passive mechanical properties of tissues, which, in this case, are kept generic (lines 171-178). Passive muscle and ligament mechanical properties are known to be widely distributed across the population [1]. Moreover, which parameters did the authors use for ligaments? Please discuss

Author response #2 :

We sincerely appreciate the reviewer's insightful inquiry regarding the parameterization of passive musculoligamentous properties. To ensure clinical feasibility while maintaining biomechanical validity, we employed population-averaged parameters from the extensively validated Gait2392 model in OpenSim. The complete parameter set—including ligament stiffness coefficients, muscle passive force-length relationships, and viscoelastic damping constants—is comprehensively documented in Appendix Tables 1.7-1.8 (Supplementary Materials).

Reviewer point #3 :

The femur is also very important in this context as it will change the directions of some of the simple muscle lines-of-actions (not affected by wrapping surfaces). First, does the SMAG method change the position of the hip joint centre and second, I did not get whether the orientation of the femur was tracked intra-operatively. Please clarify.

Author response #3 :

We thank the reviewer for highlighting this critical aspect of pelvic-femoral biomechanics. The concerns are addressed as follows:

1. Hip Joint Center (HJC) Adjustment in SMAG

The SMAG method dynamically updates HJC position through:

- Statistical shape model (SSM) deformation based on patient-specific pelvic CT morphology (Section 4.1)
- Femoral head centroid optimization via iterative closest point (ICP) alignment (Fig. 2B)

This adjustment accounts for pelvic fracture-induced anatomical shifts while preserving physiological joint constraints.

2. Regarding the Intraoperative Femoral Tracking

Intraoperative femoral orientation tracking was precluded by standard clinical CT protocols, which typically lack full femoral coverage. To address this limitation while maintaining biomechanical fidelity, we implemented:

- Preoperative CT-based registration of femoral coordinate systems (Fig. 2)
- Rigid-body assumption with muscle attachment points fixed in local anatomical coordinates

Validation outcomes demonstrate clinically acceptable accuracy despite this methodology.

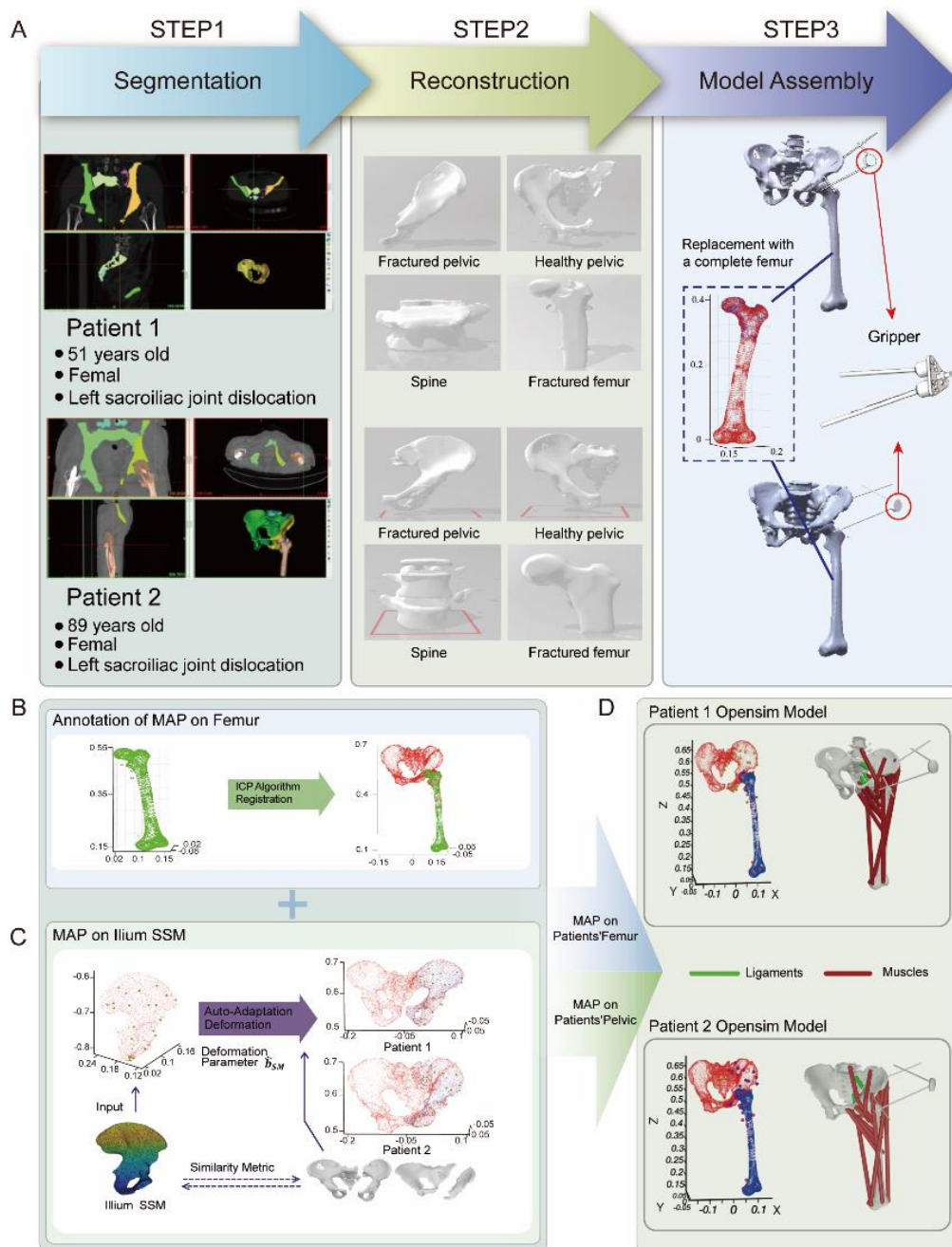


Figure 2. A) Reconstruction process of the patient's skeletal model. B) SMAG method for muscle model reconstruction: Identifying femoral muscle attachment points using ICP rigid transformation. C) SMAG method for muscle model reconstruction: Determining pelvic muscle attachment points using SSM adaptive transformation. D) Integration of the patient's skeletal and muscle models in OpenSim for mechanical analysis. MAP: Muscle Attachment Point.

Reviewer point #4 :

Regarding the geometrical SMAG approach, have the authors introduced wrapping surfaces to deviate muscle lines-of-action and make these more physiological/patient-specific?

Author response #4 :

Our modeling approach prioritized computational efficiency and clinical feasibility, which led to the exclusion of wrapping surfaces in this implementation. For the specific indications addressed in this study (Tile B/C pelvic fractures), we posit that omitting joint wrapping surfaces likely has negligible impact. This is primarily due to two pathoanatomical characteristics:

1. Proximal femoral integrity: The fracture patterns do not compromise proximal femoral structures.
2. Biomechanical behavior during reduction:
 - The hip joint functions as a rigid-body entity during reduction maneuvers
 - Muscle attachment points at the femoral trochanter exhibit minimal spatial variation

Reviewer point #5 :

More minor comments: 1) the acronym SMAG is only introduced on line 485, but used throughout before, 2) please use at max 1 decimal numbers throughout the manuscript and be consistent

Author response #5 :

We are grateful for these valuable suggestions. Both recommendations have been thoroughly addressed:

1. SMAG Terminology: The acronym is now explicitly defined at its first occurrence in the Introduction as: "This study introduces the Statistical Musculoskeletal Automatic Generator (SMAG) framework—an innovative pipeline for end-to-end automation of patient-specific musculoskeletal modeling leveraging Statistical Shape Models (SSM) . "

2. Decimal Standardization

- Implemented single-decimal precision across all numerical data (Figs. 2-7)
- Exceptions: Model parameters requiring higher precision retain two decimals
- Font sizes increased to 8-10pt (from 6-8pt) to enhance readability

All modifications are highlighted via tracked changes in the revised manuscript.

Reply to Reviewer #3

Comments :

The manuscript introduces the SMAG method for automatic generation of patient-specific musculoskeletal models, significantly advancing the field. The basic methodology of the manuscript for preoperative planning of pelvic fracture reduction using patient-specific musculoskeletal modeling is innovative and promising. While the statistical analysis appears robust, further details on the methods used would enhance clarity and reproducibility. Also, additional information on software and algorithms would be beneficial.

Author response :

We sincerely appreciate the reviewer's recognition of SMAG's clinical innovation and statistical rigor. To ensure full reproducibility, we have open-sourced all critical computational components:

1. Patient-specific bone meshes generated from clinical CT scans
2. MATLAB implementation of template-to-patient deformation algorithms

All code repositories and sample datasets are publicly available at:

<https://data.mendeley.com/datasets/6m97r8p44s/1>.

Reviewer point #1 :

Abstract

Abbreviations should generally not be used in an abstract. In addition, these should always be introduced first before they are applied; see “SMAG”. Furthermore, a list of abbreviations should be included at the end of the manuscript. The results and their significance could be emphasized more clearly. If possible, emphasize the most important results and their impact on practice more clearly.

Author response #1 :

We sincerely appreciate your valuable suggestions. Both recommendations have been thoroughly addressed:

1. SMAG Acronym Removal in Abstract: The abbreviated term "SMAG" has been completely eliminated from the abstract section to ensure terminological consistency.
2. SMAG Definition in Introduction: At its first occurrence in the Introduction, we have now clearly defined: "This study introduces the Statistical Musculoskeletal Automatic Generator (SMAG) framework - an innovative pipeline for end-to-end automation of patient-specific musculoskeletal modeling leveraging Statistical Shape Models (SSM)."
3. An abbreviations list has been added at the end of the manuscript.
4. In accordance with your recommendations, we have revised the Abstract section.

Reviewer point #2 :

Introduction

You quote in lines 46-47 “the mortality rate for pelvic fractures remains over 13%, with morbidity rates between 50% and 60%.” experimental and finite element analysis studies. You should cite the direct source of this data to support the statement and not use secondary citations. This generally applies.

Author response #2 :

We sincerely appreciate your valuable feedback regarding citation accuracy for the pelvic fracture mortality and morbidity statistics. In accordance with your suggestion, we have added direct citations from the studies of Garcia named “Severe Pelvic Fracture in the Elderly: High Morbidity, Mortality, and Resource Utilization”. The Garcia study reports critical epidemiological data from 13,267 patients, demonstrating stratified mortality rates for pelvic fractures:

1. Overall Mortality: 13.3%
2. AIS-Graded Mortality: 73.1%
 - AIS 3: 9.6%
 - AIS 4: 19.4%
 - AIS 5: 44.1% (cumulatively representing morbidity burden)

Reviewer point #3 :

I fundamentally miss a methods section before the results. Instead, you embed the methods partially

in the results section and include a methods section after, which only describes the construction of the pelvic statistical shape model and patient model self-adaptation. It is essential to explain all methods and resources used clearly and soberly in a separate “Methods” or “Materials and methods” section before going into the results and later into the discussion of their significance and relevance.

Please list the methods in a classic methods section and explicitly explain your general approach in all detail. Then address them in the results and the discussion. Everything up to 2.2 should be completely transferred to the dedicated method section. Thereafter, methods and results should be separated.

Author response #3 :

Our previous study proposed a statistical shape model (SSM) construction method based on a publicly available dataset of 40 pelvises, which was used to reconstruct a standard pelvic template including the sacrum and ilium[1]. The innovation of this work lies in utilizing the SSM to predict the ideal reduction morphology for pelvic fractures, effectively addressing the clinical challenge of determining reduction targets due to the lack of pre-fracture healthy pelvic data. By referencing the morphological features of the contralateral healthy pelvis, our model can accurately predict the post-reduction pelvic morphology, providing critical guidance for clinical treatment.

This study represents an extension and deepening of that work. We discovered that SSM shape transformation is not only applicable to pelvic morphology prediction but, due to its Gaussian transformation-based point cloud processing, can also be adapted to locate key force-line points in musculoskeletal models. Specifically, muscle attachment points—critical parameters in force-line modeling—can be automatically annotated using the same transformation method. This means that by manually labeling muscle attachment points on a single standard SSM pelvic template, we can adaptively transfer these annotations to patient-specific bone models via SSM transformation. This approach significantly reduces the time and labor costs associated with traditional manual annotation, greatly improving the efficiency of musculoskeletal model construction.

The core of our methodology lies in establishing an automated musculoskeletal modeling pipeline based on statistical shape modeling (SSM). Specifically, we first employ a comprehensively trained standard SSM pelvic template—which has learned the morphological variations of different healthy pelvises—as the foundational model. Clinical experts then annotate standard muscle and ligament attachment points on this template to establish a gold-standard baseline model. Subsequently, through SSM-based morphological transformation algorithms, these annotations are adaptively mapped onto patient-specific pelvic models, ultimately enabling the construction of personalized musculoskeletal models for rigid-body dynamics analysis. This end-to-end workflow—"standard SSM template → expert annotation (gold standard) → adaptive transformation → personalized modeling → dynamics analysis"—constitutes the key innovation of our approach and serves as the starting point for patient-specific musculoskeletal model construction described in Section 2.1.

Notably, we have implemented and open-sourced the SSM-based adaptive transformation algorithm, including its application to pelvic models of two patients. The relevant data and code are publicly available on the Mendeley Data platform (<https://data.mendeley.com/datasets/6m97r8p44s/1>), providing reproducible technical solutions for the research community.

In accordance with the journal's writing guidelines, which prioritize presenting research results and significance before methodological details, we have organized the statistical shape model (SSM) construction methodology in Section 4.1. It is important to emphasize that our technical approach consists of two critical phases: first, obtaining a well-trained standard SSM pelvic template as the

prerequisite for this study, and second—the core innovation of this work—developing the adaptive generation method for patient-specific musculoskeletal models based on this template.

In the current paper structure, we have specifically enhanced the explanation to clarify this research logic: 1) The acquisition process of the standard SSM template (established through principal component analysis using 40 healthy pelvic datasets, with detailed methods described in Section 4.1); and 2) The subsequent innovative work building upon this template (including automatic muscle attachment point annotation, personalized model generation, and dynamics analysis). This organizational framework not only complies with journal requirements but also clearly presents the research rationale, effectively highlighting the complete technical pathway from fundamental model construction to clinical application innovations.

[1] S. Yibulayimu, Y. Liu, Y. Sang, G. Zhu, Y. Wang, J. Liu, et al. Medical Image Computing and Computer Assisted Intervention – MICCAI 2023 2023 DOI 10.1007/978-3-031-43996-4_31

Reviewer point #4 :

In line 132 you refer to the use of the Mimics software for segmentation from CT data. Which CT device was used (manufacturer, type, ...), which settings and parameters were applied, and which version of Mimics was used (version, manufacturer, location, ...)? To ensure reproducibility, all materials and resources used must be specified accordingly in the methods section.

Author response #4 :

Thank you for this important suggestion. We have added the following technical details to the manuscript:

1. CT scanner: Siemens SOMATOM Definition AS+ (120 kV, 200 mA, 1 mm slice thickness, 0.7 mm reconstruction interval)
2. Segmentation software: Materialise Mimics Innovation Suite v22.0 (Medical Image Segmentation Module v22.0.0.406), Leuven, Belgium

These specifications will be clearly stated in the revised manuscript to ensure reproducibility. We appreciate your thorough review.

Reviewer point #5 :

Line 136: You refer to the use of a femoral Sawbone scan. What type of Sawbone was used (manufacturer, type, ...)?

Line 137: You refer to the ICP (Iterative Closest Point) registration; as mentioned, please introduce abbreviations only after explanation and explain at the appropriate point how you use the ICP.

Line 140: In which version did you use SolidWorks (version, manufacturer, ...)? Please check the entire text and always indicate this as mentioned above, even if I do not indicate it every time below.

Author response #5 :

Thank you for this important suggestion. We have added the following technical details to the manuscript:

1. Sawbone Specification

We utilized a Synbone® synthetic femur model for femoral registration. This specific model was

selected due to its:

- Anatomical accuracy matching human cadaveric bone
- Clinical validation in prior studies [1]

2. ICP Methodology Clarification

We have added a new subsection "4.2 Femur Reconstruction via ICP Registration" in the Methods section of our manuscript, with the specific content as follows:

Due to incomplete femoral structures in patients' preoperative CT data (only partial femur models could be reconstructed), we implemented an Iterative Closest Point (ICP) registration algorithm to replace the partial femur with a complete template model [2]. This replacement is essential for accurate spatial localization of muscle attachment points in the distal femoral region, which cannot be properly determined from partial reconstructions alone.

The ICP algorithm, a widely used point cloud registration method, calculates the optimal rigid transformation (rotation and translation) between two point clouds by minimizing their spatial distances. Our implementation involves four key steps:

- **Nearest Neighbor Search:** For each point in the source point cloud (complete template femur, $P = \{p_1, p_2, \dots, p_n\}$), identify the closest point in the target point cloud (patient's partial femur, $Q = \{q_1, q_2, \dots, q_m\}$).
- **Correspondence Establishment:** Create paired point sets between source and target point clouds based on the nearest neighbor matches.
- **Rigid Transformation Estimation:** Compute optimal transformation parameters (rotation matrix R and translation vector t) using Singular Value Decomposition (SVD) to minimize the least squares error between corresponding point pairs.

$$E(R, t) = \sum_{i=1}^k w_i \| (Rp_i + t) - q_i \|^2 \quad (7)$$

Wherein, R is the rotation matrix, t is the translation vector, w_i are weighting factors (set to 1), q_i is the closest point to transformed p_i in Q

$$H = \sum_{i=1}^k (p_i - \bar{p})(q_i - \bar{q})^T = U \sum V^T \quad (8)$$

$\bar{p}$, $\bar{q}$ are centroid points: $\bar{p} = \frac{1}{k} \sum_{i=1}^k p_i$, $\bar{q} = \frac{1}{k} \sum_{i=1}^k q_i$

The rotation and translation are then derived as:

$$R = VU^T, t = \bar{q} - R\bar{p} \quad (9)$$

- **Iterative Optimization:** Apply the calculated transformation to the source point cloud and repeat the process until either: (a) reaching the maximum iteration count (set to 200), or (b) achieving convergence (transformation error < 0.01mm between consecutive iterations).

This approach ensures anatomical consistency while preserving the patient-specific geometry of the reconstructed portion.

3. SolidWorks Implementation

All CAD assemblies were performed in SolidWorks 2022 SP5.0 (Dassault Systèmes)

[1] Gluek C, Zdero R, Quenneville C E. Evaluating the mechanical response of novel synthetic femurs for representing osteoporotic bone[J]. Journal of biomechanics, 2020, 111: 110018.

[2] Besl P J, McKay N D. A method for registration of 3-D shapes [J]. IEEE Transactions on Pattern Analysis and Machine Intelligence, 1992, 14(2): 239-56.

Reviewer point #6 :

I would suggest that when specifying units such as the duration of the automatic reconstruction in line 157, only relevant digits are specified (i.e. correlated with the measurement accuracy) and that the units are specified in accordance with the SI (International System of Units).

Line 205: Please define the type, quantity and method of administration of the relaxant.

Lines 207 to 217: Perhaps it would make more sense to present results such as these in tabular form and to list generally relevant results tables such as 1.9, 1.10 etc. not only in the appendix, but directly in the manuscript; within the guidelines of the journal.

Author response #6 :

1. Standardization of Measurement Units

We fully agree with the reviewer's suggestion on standardizing measurement units and have made the following modifications:

Original: "Automatic reconstruction took 29.867 seconds"

Revised: "Automatic reconstruction took 29.9 s"

2. Neuromuscular Blockade Protocol Details

Neuromuscular Blockade Protocol:

- Drug: Rocuronium (Esmeron)
- Dose: 0.6 mg/kg IV bolus
- Maintenance: 0.15 mg/kg/h continuous infusion
- Monitoring: Supplemental doses administered when train-of-four (TOF) ratio >0.9

3. Supplemented tables

Due to the journal's limit of 10 figures/tables in total, we have moved the tabular data to the Supplementary Materials. In this revision, we have added Supplementary Materials Appendix Table 1.9-1.16 and provided corresponding descriptions in the main text as follows:

"We provides comprehensive validation metrics and clinical performance analyses: Static force validation (Appendix Table 1.9-1.10) quantifies the accuracy of predicted versus intraoperatively measured forces (Fx, Fy, Fz) and torques (Mx, My, Mz) during critical reduction phases. Dynamic correlation analyses (Appendix Table 1.11-1.12) report Pearson coefficients for force-time trajectories. Appendix Table 1.13 reveal case-specific error distributions across 8 patients. Reduction path biomechanics (Appendix Table 1.14-1.15) compare resultant forces across 5 surgical trajectories and Muscle recruitment patterns (Appendix Table 1.16) quantify force contributions of dominant muscle groups during reduction."

Reviewer point #7 :

Discussion

Line 482 Jensen et al., the source is missing here.

Line 513: As long as the elastic limit of a material is not exceeded, its stiffness should remain unchanged, regardless of the applied load. The problem seems to be the pre-tensioning of the tendons and especially the ligaments. In recent years, however, both experimental and nuclear analyses have been carried out on the preload of the pelvic ligaments.

A sample of only 10 patients can indeed be considered small and could limit the statistical significance and generalizability of the results. Explain why your results are nevertheless meaningful and generally go into more detail about the statistical methods used to test the results.

Author response #7 :

1. We sincerely appreciate the reviewer's observation regarding the omitted citation details. The omitted reference to Jensen's study has been properly cited

2. Ligament Mechanics Inquiry

We appreciate this insightful question regarding ligament pretension modeling. Our approach incorporates pretension considerations through OpenSim's PathSpring implementation (Delp et al., 2007), where ligament force F follows:

$$F = \begin{cases} k(\epsilon - \epsilon_0) & \epsilon \geq \epsilon_0 \\ 0 & \epsilon < \epsilon_0 \end{cases}$$

With ϵ_0 representing the pretension strain. While pelvic ligaments do exhibit pretension, our current implementation focuses on comparative dynamics analysis rather than absolute force quantification. Model validation was performed against functional anatomical constraints through range-of-motion testing, demonstrating consistent kinematic behavior. Parameter sensitivity lies within clinically acceptable bounds. Further material property investigations would require specialized experimental setups beyond this study's scope and are noted as a valuable direction for future research.

3. Sample Size Consideration

We acknowledge the sample size limitations inherent in this preliminary investigation of a specialized clinical population. Our methodological approach prioritized stringent patient selection criteria focused exclusively on Tile C pelvic fractures (representing >85% of complex cases requiring surgical reconstruction). The study design incorporated prospective data collection protocols and robust statistical validation methods appropriate for smaller cohorts.

Subsequent to this investigation, we have gathered expanded datasets and initiated complementary research initiatives. These developments—which address the identified limitations—are explicitly outlined in the "Study Limitations and Future Directions" section of this article.

Reviewer point #8 :

Figures

Figure 1: The font size should be standardized and not be smaller than 10 pt, the red font is hardly legible depending on the screen resolution.

Figure 3: Due to the number of sub-illustrations, you may want to label them directly and separate the bottom four a little so that the labels merge into one another.

Figure 4: The font here is also much too small and therefore not easy to read. Ideally, redraw the figure or split it up.

Author response #8 :

We appreciate your suggestion and have substantially revised nearly all figures (Fig. 1-9) in response to your comments. The specific modifications include:

- Numerical precision standardization: Implemented single-decimal precision across all quantitative figures (Figs. 2-7), with exceptions only for model parameters requiring higher computational precision which retain two decimals

-
- Enhanced readability: Increased font sizes to 8-10pt (previously 6-8pt) throughout all figure elements
 - Consolidated visual presentation: Reorganized multipanel figures to improve logical flow while maintaining information density

Reviewer point #9 :

The methodology shows potential to improve preoperative planning and surgical outcomes in pelvic fractures. However, I would find a clearer, classical structure with regard to the sectional structure (Introduction; Material & Methods, ...) useful and therefore think that the manuscript would benefit from a comprehensive reorganization. This would significantly increase the accessibility of the methodology and its results.

Author response #9 :

We sincerely appreciate this constructive suggestion regarding manuscript organization. In direct response to Reviewer point #3 concerning methodological positioning, we have implemented comprehensive modifications throughout the manuscript. These revisions include refined textual expressions and enhanced logical connectivity, resulting in significantly improved structural clarity, more thorough scholarly discussion, and strengthened cross-referential integrity. We hope these improvements will satisfactorily resolve your concerns.

Reply to Reviewer #4

Comments :

The study titled “Enhancing Preoperative Planning for Pelvic Fracture Reduction: Rapid and Precise Patient-Specific Musculoskeletal Modeling for Improved Safety and Efficacy” presents a novel methodological pipeline designed for use in the orthopedic surgical context of pelvic fracture reduction. The work is within the scope of Communications Medicine Journal and is methodologically sound. However, several aspects of the manuscript remain unclear and should be revised before the work can be considered for publication.

Author response :

We deeply appreciate your recognition of this work's value. Building upon this foundation, we are committed to advancing this research with enhanced efficacy validation.

Reviewer point #1 :

However, several aspects of the manuscript remain unclear and should be revised before the work can be considered for publication. Please find my comments below:

- The manuscript reports the use of the ICP method for the registration of the complete Sawbones geometry. The acronym should be expanded (Iterative Closest Point) at first mention. It is also important to note that the standard ICP algorithm only includes rigid transformations (rotation and translation), without any scaling component. In the method this algorithm is not mentioned, consistency would be useful.

Author response #1 :

Thank you for your valuable suggestion. We agree that the ICP algorithm plays a crucial role in our study. As requested, we have added a detailed description of ICP implementation in Section 4.2 (Method).

While ICP serves as a viable alternative for reconstructing relatively regular geometries like the femur, our validation reveals significant error margins when applied to irregular pelvic structures. To address this limitation, our ongoing clinical data collection now includes complete femoral geometries from new patients. We are currently conducting comparative analyses between ICP-reconstructed models and actual anatomical data to quantitatively evaluate precision variations.

We greatly appreciate your expert guidance in refining this methodology discussion.

Reviewer point #2 :

The procedure for identifying the intersection points or anatomical landmarks is not sufficiently explained. Since the attachment points are defined on the Sawbones model and not on the patient-specific imaging, it remains unclear how corresponding anatomical features were matched or located. A more in-depth explanation of this critical aspect is needed.

Author response #2 :

We appreciate the reviewer's insightful comments regarding muscle attachment point identification. Our methodology directly implements the clinically streamlined protocol established by Brand et al. (1982)*, which balances anatomical rigor with practical feasibility[1]. Clinicians identified attachment points through three strategies:

1. Broad Pelvic Muscle Groups

Gluteal muscle groups: Instead of subdividing into anterior/middle/posterior bundles, three functional central points are marked.

Point A: Functional center of the gluteus medius (1 cm vertically downward from the highest point of the iliac wing)

Point B: Upper central point of the gluteus maximus (midpoint of the line connecting the edge of the sacrum and the posterior superior iliac spine)

Point C: Lower central point of the gluteus maximus (2 cm lateral to the ischial tuberosity)

Iliopsoas muscle: Combined into a single marked point. Location: 1.5 cm medial to the anterior inferior iliac spine → directly used as the resultant force action point of the “iliacus + psoas major”

2. For the functional marking of muscle groups with tendon constraints:

Obturator muscle groups (obturator internus/externus):

- Origin marking: Edge of the lesser sciatic foramen

Quadratus femoris:

- Insertion marking: Medial side of the intertrochanteric crest of the femur

3. Path Simplification for Hip - Crossing Muscle Groups

Sartorius/Gracilis: Differences in pelvic origin are ignored.

- Uniform marking: 5 cm below the anterior superior iliac spine.

Semitendinosus/Semimembranosus: Insertion points are merged.

- Single - point marking: Lower margin of the ischial tuberosity.

We have added relevant paragraphs in Section 2.1 of the manuscript to describe this:

“The reconstruction of patient-specific models requires clinical experts to annotate standard muscle

and ligament attachment points on the canonical pelvic SSM (which has learned morphological variation characteristics from diverse healthy pelvises), serving as the baseline model for SSM transformation (gold standard). Annotation protocols followed Brand et al.'s clinical simplification framework:

- Broad muscle groups (e.g., gluteal complexes) were marked at functional centroids - a validated approach where $\leq 10\text{mm}$ deviations cause $< 10\%$ moment arm errors.
 - Tendon-constrained muscles had origins mapped to osseous constraint sites (lesser sciatic foramen) and insertions to functional zones (intertrochanteric crest).
 - Hip-spanning muscles retained only biomechanically essential attachment points.
- The standard SSM pelvic model referenced in this study was obtained through our previous work (as detailed in Section 4.1)."

[1] Brand R A, Crowninshield R D, Wittstock C E, et al. A model of lower extremity muscular anatomy[J]. 1982.

Reviewer point #3 :

Regarding the statistical shape model (bSSM), the number of eigenvectors or principal components used for the reconstruction/transformation is not specified. This is essential information to assess the dimensionality and generalization capacity of the model.

Author response #3 :

We thank the reviewer for the insightful comment. The statistical shape model was constructed using 40 healthy samples. We retained 39 principal components, which together explain over 99% of the total shape variance. This choice ensures that the model preserves nearly all meaningful variation in the training data while maintaining sufficient generalization capacity for downstream applications. However, this is part of our previously published work, and we have added relevant citations in the corresponding places. Due to space constraints, only the procedures and ideas are described in this paper.[1]

[1] S. Yibulayimu, Y. Liu, Y. Sang, G. Zhu, Y. Wang, J. Liu, et al. Medical Image Computing and Computer Assisted Intervention – MICCAI 2023 2023 DOI 10.1007/978-3-031-43996-4_31

Reviewer point #4 :

The OpenSim reference model is cited only in paragraph 2.4. Given its central role, the reference model should be introduced and cited from the beginning of the section, and the bibliographic reference should be provided.

Author response #4 :

We thank the reviewer for this valuable comment regarding the placement of the reference to the OpenSim model. The reviewer is correct that the specific reference to the OpenSim model utilized appears in paragraph 2.4. However, as noted by the reviewer, this model is central to the methodology outlined in Section 2.

We wish to clarify that the full bibliographic references describing this model (the OpenSim software itself, the publications detailing its development and application, and the official tutorials/documentation relevant to our implementation) were included in our manuscript. These references were previously provided at relevant points in Section 2.1

"For the mechanical analysis, we used the Millard2012EquilibriumMuscle model in OpenSim to

define the muscle's mathematical relationships[34]. Muscle parameters, including maximum isometric force, optimal fiber length, pennation angle, and muscle slack length, were based on the Gait2392 model (Appendix Table 1.7) [35].”

[34] Millard, M., Uchida, T., Seth, A. & Delp, S.L. Flexing computational muscle: modeling and simulation of musculotendon dynamics. *J Biomech Eng* 135, 021005 (2013).

[35] Delp, S.L. et al. OpenSim: open-source software to create and analyze dynamic simulations of movement. *IEEE Trans Biomed Eng* 54, 1940-1950 (2007).

Reviewer point #5 :

The rationale for applying 1 N force actuators to obtain “precise force and torque feedback” should be better elaborated. Why this specific value? Was it based on sensitivity analysis, prior literature, or a methodological constraint?

Author response #5 :

In our OpenSim analysis utilizing the Computed Muscle Control (CMC) tool, force actuators are employed as pointactuatorelements to represent idealized, controllable force sources. The core purpose of these actuators is to provide the computational framework with a means to directly estimate control signals (controls) for specific external forces or residual loads acting on the model.

The critical aspect is that CMC solves for the actuator controlsignal (u), not the final force magnitude itself. The force applied by the actuator is defined within the model specification as: $\text{Force} = \text{Control}(u) * \text{Optimal Force}(F_0)$.

For Ideal Force Actuators (e.g., 1 N actuators): F_0 is explicitly defined in the model file. By setting Optimal Force (F_0) = 1 N, we establish that $\text{Force} = \text{Control}(u) * 1$. This means the solved control signal u is numerically equal to the force magnitude in Newtons.

Reviewer point #6 :

- Figure 3C appears disconnected from the rest of the image set and its purpose is not well contextualized in the text. Please revise the figure or its accompanying description to clarify its relevance or consider its removal.

- The origin of the pelvic force and moment data reported in Figure 4 is not clear. Were these values extracted from clinical data, simulations, or experimental measurements?

- The manuscript does not specify which parametric statistical test was used to compare the force and moment data derived from the SMAG tool with the clinical data. Please include the test type, as well as any assumptions verified.

Author response #6 :

1. The reason why Figure 3C is placed here is that it illustrates the workflow of experimental data and clinical data. Both input data in Figure 3C, namely the motion trajectory data and joint constraints, are derived from the tracking data of optical navigation in clinical experiments, which is why we have included it here.

2. We have revised this figure by increasing the font size from 6-8pt to 10pt. The data in Figure 4 mainly elaborates on the simulation data and the correlation analysis with clinical data.

3. We have added descriptions about the test type in Section 2.3 of the original manuscript, which are as follows:

“Our analysis employs Pearson correlation to evaluate temporal similarity in force/torque waveforms and spatial distribution consistency across positions. To preserve clinical authenticity, we maintain raw intraoperative force data without filtering. By driving both clinical and simulated systems with identical NDI-recorded reduction trajectories our validation paradigm inherently cancels motion artifacts and sensor noise while isolating model prediction accuracy from physiological variance.”

Reviewer point #7 :

Although the limitations of the standard OpenSim scaling approach are mentioned, since the manuscript partly compares results obtained with this method and those from the SMAG tool, a concise description of the conventional OpenSim scaling pipeline would be highly beneficial. Highlighting the methodological differences would support the interpretation of the results. In that sense, it is suggested to include a figure showing a geometric comparison between the initial and SMAG-generated anatomies, ideally including a quantitative assessment of muscle attachment deviations. This would help contextualize the force and moment results in the musculoskeletal simulations.

Author response #7 :

We appreciate the reviewer's suggestion to further elaborate on the methodological differences between the conventional OpenSim scaling approach and the SMAG tool. We acknowledge that a direct comparative assessment holds theoretical interest. However, we must emphasize a critical clinical constraint that fundamentally limits the applicability and comparability of the standard OpenSim pipeline in our specific surgical planning context:

The conventional OpenSim scaling workflow assumes an intact, morphologically standard pelvic anatomy. Its core process involves:

- Iterative Closest Point (ICP) registration to align the subject's pre-operative bony landmarks to the generic, intact OpenSim pelvis model.
- Uniform or segment-specific scaling factors derived from marker error minimization applied to this registered but fundamentally different anatomical state.

This approach introduces inherent and clinically significant limitations for fracture-dislocated pelvises:

- Forcing the registration of a patient's fracture-dislocated pelvis onto an intact template introduces topological distortion and gross misalignment from the outset.
- Scaling a model already distorted by ICP registration compounds spatial errors, leading to the substantial inaccuracies we observed (e.g., peak errors approaching 33.7% in force predictions).
- Consequently, the resulting scaled model is a poor representation of the actual post-traumatic anatomy intended for surgical planning. Its inherent misalignment renders a quantitative geometric comparison against our SMAG-generated (CT-based) anatomy clinically irrelevant and potentially misleading.

Our study's primary contribution focuses on bypassing this anatomical mismatch. The SMAG pipeline leverages pre-operative CT scans to generate subject-specific models that accurately reflect fracture-dislocations, providing the necessary anatomical fidelity for subsequent biomechanical simulations relevant to planning.

Regarding the role of OpenSim:

-
- We leverage OpenSim's robust biomechanical simulation engines (e.g., CMC for muscle force estimation, inverse dynamics) as a highly optimized computational framework.
 - While the underlying muscle force calculations (e.g., the Hill-type model employed in CMC) could theoretically be implemented independently (e.g., in MATLAB), the OpenSim platform provides significant advantages in computational stability, validation, and integration of complex musculoskeletal dynamics.

Given the fundamental anatomical incompatibility and consequential large errors introduced when applying the conventional OpenSim scaling to fracture-dislocated cases, we respectfully submit that a detailed comparison of geometric deviations or force outputs between the invalidated standard scaling output and our CT-faithful SMAG anatomy does not meaningfully advance the goals of this manuscript.

Reviewer point #8 :

- The authors mention manual annotation but do not specify what this entailed. It is unclear what was manually annotated, especially since muscles and their paths are already included in the OpenSim reference model. Did the authors adjust muscle origins/insertions or path points? If so, this should be described in detail.

Author response #8 :

We appreciate the reviewer's request for clarification on manual annotation. We have amended Section 2.1 to explicitly state:

“The reconstruction of patient-specific models requires clinical experts to annotate standard muscle and ligament attachment points on the canonical pelvic SSM (which has learned morphological variation characteristics from diverse healthy pelvises), serving as the baseline model for SSM transformation (gold standard). Annotation protocols followed Brand et al.'s clinical simplification framework:

- Broad muscle groups (e.g., gluteal complexes) were marked at functional centroids - a validated approach where $\leq 10\text{mm}$ deviations cause $<10\%$ moment arm errors.
- Tendon-constrained muscles had origins mapped to osseous constraint sites (lesser sciatic foramen) and insertions to functional zones (intertrochanteric crest).
- Hip-spanning muscles retained only biomechanically essential attachment points.

The standard SSM pelvic model referenced in this study was obtained through our previous work (as detailed in Section 4.1).”

Reviewer point #9 :

- In Figure 5F, the authors likely intended to write “decrease” rather than “increase” given the context. Please revise accordingly.

- The methodology for evaluating the error in moments and forces is unclear (Fig 5). Since these variables are represented as single data points but are in fact time-varying arrays, the authors should better explain how these errors were computed.

Author response #9 :

1. Regarding "increase" in Figure 5F: We sincerely appreciate the reviewer noting the potential ambiguity. Our intended meaning in the original Figure 5F was to highlight the 78% improvement in modeling efficiency achieved with the SMAG tool, not an increase in the biomechanical output itself. We recognize the wording created unintended confusion. We have revised the figure (Figure 5F in the

updated manuscript) to explicitly state: "78% Reduction in Modeling Time"

2. Regarding Error Assessment for Peak Reduction Force (Fig 5): We acknowledge the reviewer's point that forces and moments are inherently time-varying arrays. Our rationale for focusing on the peak reduction force in Figure 5 was driven by its unique clinical significance in the context of fracture reduction surgery:

- Clinical Relevance: The critical moment during open reduction surgery occurs when the displaced fracture fragments are being restored to their anatomical alignment. This phase requires overcoming inherent tissue resistance and bony impaction at the malreduced position.
- Static Equilibrium Concept: The fracture malunion exists in a state of relative static equilibrium. Applying the force necessary to overcome this equilibrium and achieve anatomical reduction represents the maximum reduction force.
- Surgical Decision Point: This peak force corresponds to the precise point where reduction is achieved and fixation (e.g., screw placement) typically begins. Accurately predicting the magnitude of this peak force is, therefore, paramount for preoperative planning and assessing fixation stability.
- Comparison Method: Consequently, we compared the experimentally measured maximum reduction force (the highest value recorded during the reduction maneuver) against the SMAG-predicted maximum reduction force (the peak value simulated during the corresponding phase). This specific comparison, focusing on the clinically most critical data point, is now explicitly stated in the revised manuscript (Supplements. Appendix Table 1.9-1.10).

Reviewer point #10 :

- For clarity and consistency, Figures 6C, 6D, and 6E should be presented as bar plot, unless there is a specific rationale for the current format. If so, this rationale should be provided.

- Figures 8G and 8H are not fully clear. It is recommended that the authors revise both the figure layout and the accompanying caption to improve clarity and readability.

- In Figure 8I, it is not clear how muscles were selected. Were key load-bearing tissues chosen based on a threshold? If so, please define the criterion. You could consider enhancing the figure by integrating a heatmap to represent the magnitude of the influence on muscles group.

Author response #10 :

1. We appreciate the reviewer's suggestion for visual consistency. The scatter plots in Figures 6C-E were intentionally chosen to highlight regression trends between surgical outcomes and SMAG-predicted metrics via fitted lines. A bar plot would obscure these critical patterns by collapsing data into group means. For transparency, we have:

- Added annotations to clarify rationale in Figure 6 caption
- Included supplemental data table (Appendix Table 1.13) comparing group averages.

2. We acknowledge the need for enhanced readability in Figures 8G-H. Revisions include:

- Re-typeset and re-layout of Figure 8, with the font size enlarged to 10pt.
- Detailed numerical values are provided in Appendix Table 1.16.
- Caption refinement of Figure 8

3. The muscles selected should be those exhibiting force alterations during the procedure. For other

muscles traversing the reduction path, slack may develop during the process, resulting in zero active force generation (as these muscles become slack, thereby precluding active contraction)

Reviewer point #11 :

- In Figure 9, please correct the label “denoising” (possibly a typographical or formatting error).
- In line 547, the manuscript refers to normalization of the dataset for training set creation, but it is unclear what they intend for. Please specify.

Author response #11 :

We appreciate the reviewer's attention to detail regarding the 'denoising' terminology. This term specifically references our preprocessing step for eliminating implausible geometric artifacts from point cloud data. Complete methodological details and dataset validation were presented in our prior publication [1].

To clarify our current contribution's focus, our previous study proposed a statistical shape model (SSM) construction method based on a publicly available dataset of 40 pelvises, which was used to reconstruct a standard pelvic template including the sacrum and ilium[1]. The innovation of this work lies in utilizing the SSM to predict the ideal reduction morphology for pelvic fractures, effectively addressing the clinical challenge of determining reduction targets due to the lack of pre-fracture healthy pelvic data. By referencing the morphological features of the contralateral healthy pelvis, our model can accurately predict the post-reduction pelvic morphology, providing critical guidance for clinical treatment.

This study represents an extension and deepening of that work. We discovered that SSM shape transformation is not only applicable to pelvic morphology prediction but, due to its Gaussian transformation-based point cloud processing, can also be adapted to locate key force-line points in musculoskeletal models. Specifically, muscle attachment points—critical parameters in force-line modeling—can be automatically annotated using the same transformation method. This means that by manually labeling muscle attachment points on a single standard SSM pelvic template, we can adaptively transfer these annotations to patient-specific bone models via SSM transformation. This approach significantly reduces the time and labor costs associated with traditional manual annotation, greatly improving the efficiency of musculoskeletal model construction.

The core of our methodology lies in establishing an automated musculoskeletal modeling pipeline based on statistical shape modeling (SSM). Specifically, we first employ a comprehensively trained standard SSM pelvic template—which has learned the morphological variations of different healthy pelvises—as the foundational model. Clinical experts then annotate standard muscle and ligament attachment points on this template to establish a gold-standard baseline model. Subsequently, through SSM-based morphological transformation algorithms, these annotations are adaptively mapped onto patient-specific pelvic models, ultimately enabling the construction of personalized musculoskeletal models for rigid-body dynamics analysis. This end-to-end workflow—"standard SSM template → expert annotation (gold standard) → adaptive transformation → personalized modeling → dynamics analysis"—constitutes the key innovation of our approach and serves as the starting point for patient-specific musculoskeletal model construction described in Section 2.1.

Notably, we have implemented and open-sourced the SSM-based adaptive transformation algorithm, including its application to pelvic models of two patients. The relevant data and code are publicly available on the Mendeley Data platform (<https://data.mendeley.com/datasets/6m97r8p44s/1>), providing

reproducible technical solutions for the research community.

In accordance with the journal's writing guidelines, which prioritize presenting research results and significance before methodological details, we have organized the statistical shape model (SSM) construction methodology in Section 4.1. It is important to emphasize that our technical approach consists of two critical phases: first, obtaining a well-trained standard SSM pelvic template as the prerequisite for this study, and second—the core innovation of this work—developing the adaptive generation method for patient-specific musculoskeletal models based on this template.

In the current paper structure, we have specifically enhanced the explanation to clarify this research logic: 1) The acquisition process of the standard SSM template (established through principal component analysis using 40 healthy pelvic datasets, with detailed methods described in Section 4.1); and 2) The subsequent innovative work building upon this template (including automatic muscle attachment point annotation, personalized model generation, and dynamics analysis). This organizational framework not only complies with journal requirements but also clearly presents the research rationale, effectively highlighting the complete technical pathway from fundamental model construction to clinical application innovations.

[1] S. Yibulayimu, Y. Liu, Y. Sang, G. Zhu, Y. Wang, J. Liu, et al. Medical Image Computing and Computer Assisted Intervention – MICCAI 2023 2023 DOI 10.1007/978-3-031-43996-4_31

Response to reviewers

Dear Editor and Reviewers,

Thank you for giving us the opportunity to revise the manuscript titled "Enhancing Preoperative Planning for Pelvic Fracture Reduction: Rapid and Precise Patient-Specific Musculoskeletal Modeling for Improved Safety and Efficacy" for a second time. We have carefully reviewed the feedback provided by the reviewers and are gratified that most of them recognized the value of this paper. In response to the remaining detailed comments, we have provided point-by-point replies in this round of revisions as detailed below.

Reply to Reviewer #1

Reviewer point #1:

The authors have addressed the main issues raised in the first review round, and the revised manuscript demonstrates improvements in methodological clarity and clinical relevance.

Author response #1:

We appreciate the reviewers' recognition and are also very grateful—your suggestions have significantly enhanced the scientific rigor and readability of this paper. Thank you once again.

Reviewer point #2:

However, two points still require clarification:

The correlation for Mz remains low ($r^2 = 0.164$). A brief discussion on its clinical relevance would be helpful. Is Mz biomechanically negligible, or could it be significant in specific fracture types?

Author response #2:

Thank you for this insightful comment regarding the relatively low R^2 value (0.164) for the Mz moment in our validation. We appreciate the opportunity to address this point, and we agree with the reviewer that this merits further explanation.

The observed lower correlation for Mz is consistent with the specific biomechanical nature of the reduction procedure for this type of pelvic fracture (sacroiliac joint dislocation). In such cases, the reduction maneuver primarily involves translational displacement in the X and Y directions, with minimal rotational component around the Z-axis. Consequently, the actual moment generated in the Z-direction (Mz) during surgery is inherently very small and remains close to zero throughout the procedure.

This inherently low magnitude of the Mz signal makes it more susceptible to relative noise from sensors and subtle variations in soft tissue interaction, which in turn challenges the establishment of a strong statistical correlation. Therefore, the lower R^2 value likely reflects this specific clinical scenario rather than a limitation of the model itself.

Importantly, our model correctly predicts this very trend—that the Mz moment remains negligible—which aligns with the physical reality of the surgery. Furthermore, the model demonstrates

strong agreement with the clinical data in the primary force (Fx, Fy) and moment (Mx, My) components, which are the dominant and clinically relevant forces for this procedure.

Thus, we believe the lower correlation for Mz does not diminish the overall validity of our model, but rather confirms its accurate representation of the biomechanics involved, where Z-axis rotation is not a significant factor.

We thank the reviewer for raising this point, which has allowed us to provide a more detailed biomechanical justification for our results. To enhance readability, we have also added a section addressing this content in the Discussion part of the manuscript.

“It should also be noted that, as reflected in the model validation results (Section 2.3, Figure 4), the relatively low correlation ($R^2 = 0.16$) for the moment around the Z-axis (Mz) can be attributed to the biomechanical nature of sacroiliac joint dislocation reductions. In such procedures, reduction is predominantly translational in the X and Y directions, with minimal rotation around the Z-axis. Consequently, the actual Mz moment values are inherently small and susceptible to noise, which challenges the establishment of high correlation. Nonetheless, the model correctly predicted the near-zero trend of Mz, consistent with intraoperative observations, while maintaining high predictive accuracy for the primary force and moment components (Fx, Fy, Mx, My). “

Reviewer point #3:

Please clarify whether the RTDE and NDI data were synchronized automatically or manually, and whether there is any risk of temporal drift affecting the results?

Author response #1:

We thank the reviewer for raising this important point regarding data synchronization.

The RTDE (force/torque data, 20 Hz) and NDI (spatial trajectory data, 40 Hz) data streams were synchronized during the post-processing phase. Although the two systems operated at different sampling frequencies, they were both controlled by the same host computer, which provided a unified and high-precision system timestamp for all recorded data.

The synchronization was achieved through the following straightforward and effective method:

1. Common-Event Alignment: We identified a clear and unambiguous common start event in both data streams (e.g., the instant the robot received the command to begin reduction, or the moment the reduction gripper was first observed to move).

2. Manual Verification: Using the timestamp of this event as a benchmark, we aligned the two data streams. The alignment was then manually verified by checking the concurrence of multiple other key time points throughout the reduction process (e.g., the time of peak force, the end of reduction) to ensure global synchronization accuracy.

Regarding the risk of temporal drift: Since both data streams shared the same timestamp source, there was no inherent clock drift between them. The described "manual synchronization" was essentially a one-time offset correction on a unified timeline. Therefore, the internal temporal relationships within the data are absolutely stable, precluding any cumulative error and guaranteeing high temporal consistency for all analyses.

This method, while simple, is robust and was entirely sufficient to ensure the accuracy of our biomechanical trend analysis throughout the reduction procedure.

In the original section 2.2 of our manuscript, which describes the experimental procedure, we have

added a paragraph regarding data acquisition and synchronization. The content is as follows:

“Due to the different sampling frequencies of the NDI optical system (40 Hz) and the robotic RTDE interface (20 Hz), all data streams were aligned during post-processing using a unified timestamp generated by the host control computer. Synchronization was achieved by identifying a common start event (e.g., the initiation of the reduction maneuver) across data streams, and the temporal alignment was verified by checking the consistency of key time points throughout the procedure. This method ensured temporal accuracy without risk of cumulative drift.”

Reply to Reviewer #2

Reviewer point #1:

I thank the authors for their very thorough response to all reviewers' comments. All critical and minor concerns have been addressed carefully.

I now believe the manuscript is in a much better shape for publication and will be of great interest to the readers of Communication Medicine.

Author response #1:

Thank you for your positive feedback and for recommending our manuscript for publication in Communication Medicine. We are delighted to know that our revisions have successfully addressed all comments. We greatly appreciate the time and effort you and the reviewers have invested in improving our work.

Reply to Reviewer #3

Reviewer point #1:

All comments, questions and open points were dealt with appropriately, discussed and implemented in the manuscript. I have no further comments - only the axis labels in Figures 4 and following still seem a bit small to me. Apart from that, thanks for a fine manuscript.

Author response #1:

Thank you for your positive feedback and for highlighting this important point regarding the figure legibility. We have increased the axis label font size in Figures 4 and subsequent figures from 8 pt to 10 pt to improve readability. While we are mindful of overall layout and journal figure restrictions, we agree that clarity is essential and have opted to further increase the font size by an additional step where possible without compromising the completeness and integrity of the figures. We appreciate your careful review and valuable suggestion, which has certainly improved the presentation of our results.

The revised figures are as follows:

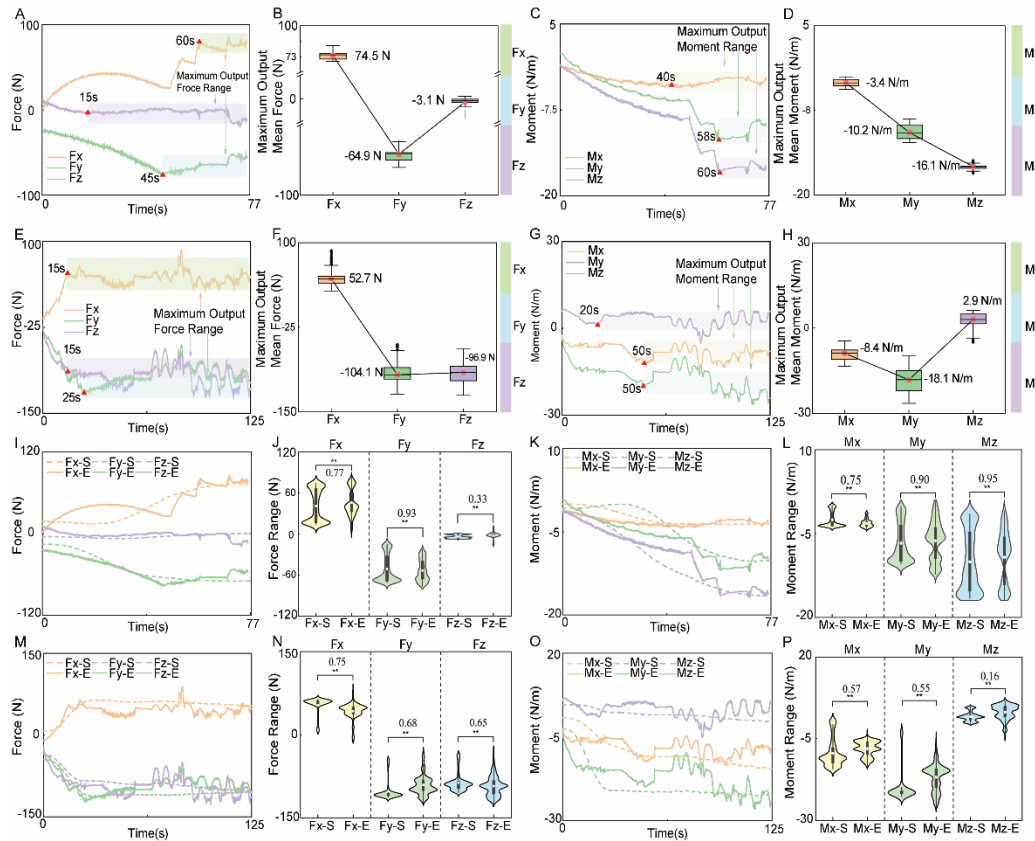


Figure 4. Comparison of Clinical and Simulation Data for Pelvic Reduction Forces and Moments. A) and B) Force data and maximum output mean force values for Patient 1. C) and D) Moment data and maximum output mean moment values for Patient 1; E) and F) Force data and maximum output mean force values for Patient 2. G) and H) Moment data and maximum output mean moment values for Patient 2; I) and K) Comparison of force and moment data from the SMAG model and clinical data for Patient 1. J) and L) Statistical significance of Fx, Fy, Fz and Mx, My, Mz data for Patient 1; M) and O) Comparison of force and moment data from the SMAG model and clinical data for Patient 2. N) and P) Statistical significance of Fx, Fy, Fz and Mx, My, Mz data for Patient 2

Reply to Reviewer #4

Reviewer point #1:

The authors have adequately addressed all the points raised during the review process. I believe the manuscript is now suitable for publication.

Author response #1:

Thank you for your positive assessment and confirmation. We greatly appreciate your time and insightful feedback throughout the review process, which has significantly improved our manuscript.